

A REPORT OF SEMESTER INDUSTRIAL TRAINING

at

QODEMATRIX

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENT FOR THE AWARD

OF THE DEGREE OF

BACHELOR OF TECHNOLOGY

(Computer Science and Engineering)



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CERTIFICATE BY COMPANY



I.K. GUJRAL PUNJAB TECHNICAL UNIVERSITY, KAPURTHALA

CANDIDATE'S DECLARATION

I, AMANJEET SINGH, hereby declare that I have undertaken semester industrial training at QODEMATRIX during the period from January 2026 to June 2026 in partial fulfillment of the requirements for the award of the degree of B. Tech (Computer Science and Engineering) at I.K. Gujral Punjab Technical University, Kapurthala.

The work being presented in this training report titled " Customer Shopping Behavior Analysis "submitted to the Department of Computer Science and Engineering at I.K. Gujral Punjab Technical University, Kapurthala is an authentic record of my individual training work carried out under the guidance and supervision provided during the training period.

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The six months industrial training Viva–Voce Examination of _____ has been held on _____ and accepted.

Signature of Internal Examiner

Signature of External Examiner

ABSTRACT

Data has become one of the most valuable assets for modern organizations, and the ability to convert raw transactional data into actionable business insight is now a core skill expected of any technology professional. This report documents the semester industrial training undertaken on the project which involved the end-to-end analysis of a retail sales transaction dataset comprising 5,901-line items across 3,003 unique orders.

The training combined three complementary tool-sets: Microsoft Excel for data cleaning, exploratory analysis and pivot-based reporting; MySQL for structured querying and aggregation of transactional records; and Microsoft Power BI for the design of an interactive, multi-page business intelligence dashboard. The dataset captured order details, shipping information, customer segments, product hierarchy (Category and Sub-Category), regional distribution, sales, profit, profit margin, cost of goods, payment mode and returns.

The analysis revealed that the organization generated total sales of approximately USD 1,565,804 and a total profit of approximately USD 152,834, representing an overall profit margin of 9.76%. The Office Supplies category contributed the highest profit, while the Furniture category, and specifically the Tables sub-category, was found to be loss-making. The West region recorded the highest profit margin (11.83%), while the Central region recorded the lowest (5.61%). Several sub-categories, including Tables, Supplies and Bookcases, were identified as net loss-making and are highlighted for managerial attention.

SQL queries were used to compute profit by sub-category, profit margin percentage by region, top-five products by profit, region-wise filters, profit by category and by state, and to isolate loss-making products. These query results were validated against the Excel pivot analysis and visualized through an interactive Power BI dashboard with slicers for Region, Category and Segment, enabling stakeholders to drill down into the data dynamically.

The training enabled the development of practical skills in data cleaning, descriptive statistics, SQL query writing, dashboard design, and the translation of analytical findings into business recommendations. The report concludes with key findings, business recommendations for improving profitability in loss-making segments, and the scope for future enhancement of the dashboard using predictive analytics.

ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to QodeMatrix Solutions for providing me with the valuable opportunity to gain practical knowledge and industry exposure in the field of Data Analytics.

I extend my sincere thanks to the entire team at QodeMatrix Solutions, especially the trainers and mentors, for their constant support, expert guidance, and effective teaching methodologies. Their in-depth knowledge, approachable nature, and dedication played a crucial role in helping me understand various Data Analytics concepts and tools such as Microsoft Excel, SQL, Power BI.

I am also thankful to the mentors for providing practical exposure through real-world projects, assignments, and case studies that helped me develop analytical thinking and problem-solving skills. Their continuous encouragement and valuable feedback significantly contributed to my learning and professional growth.

This training experience at QodeMatrix Solutions has not only enhanced my technical knowledge but has also improved my confidence, communication skills, and understanding of industry practices. The well-structured training program, hands-on sessions, and project-based learning environment provided me with valuable insights into the field of Data Analytics and Business Intelligence.

I sincerely thank QodeMatrix Solutions for playing a significant role in my professional development and for guiding me toward a successful career in Data Analytics.

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ABOUT THE COMPANY



QodeMatrix Tech Solutions Pvt. Ltd. is a rapidly growing IT services and technology company focused on delivering innovative digital solutions and industry-oriented training programs. The company specializes in software development, web development, mobile application development, cloud solutions, artificial intelligence, cybersecurity, data analytics, and business intelligence services. Its mission is to help businesses embrace digital transformation while equipping students and professionals with practical industry skills.

The organization provides customized technology solutions designed to meet the unique requirements of businesses across various sectors. Its expertise includes web application development, mobile app development, custom software solutions, UI/UX design, cloud infrastructure, and data-driven business solutions. By combining modern technologies with industry best practices, QodeMatrix aims to deliver scalable, secure, and high-performance solutions.

Apart from software services, QodeMatrix Tech Solutions is committed to skill development through industrial training and internship programs. The company offers practical learning opportunities in domains such as Data Analytics, Full Stack Development, Artificial Intelligence & Machine Learning, Mobile App Development, and Cybersecurity. These programs focus on hands-on learning, real-world projects, and industry exposure, helping students bridge the gap between academic education and professional requirements.

The company promotes a project-based learning approach where trainees work on real datasets, business scenarios, and development projects. This methodology helps participants strengthen their technical skills, analytical thinking, problem-solving abilities, and understanding of industry practices. The training programs are designed to enhance employability and prepare students for successful careers in the IT sector.

QodeMatrix Tech Solutions is supported by experienced professionals who provide mentorship, technical guidance, and practical insights throughout the learning process. The company emphasizes continuous learning, innovation, teamwork, and professional growth.

Head Office: Gurugram, Haryana

Operational Office: Mohali, Punjab

Industry: IT Services, Software Development, Training & Consulting

Key Areas: Data Analytics, Web Development, Mobile App Development, AI/ML, Cybersecurity, Cloud Solutions, Business Intelligence

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DEFINITIONS, ACRONYMS AND ABBREVIATIONS

Term	Definition
SQL	Structured Query Language, used to query and manage relational databases.
CSV	Comma-Separated Values, a plain text file format for storing tabular data.
PBIX	Power BI file extension used to save reports and dashboards.
BI	Business Intelligence, technology for analyzing and presenting business data.
KPI	Key Performance Indicator, a measurable value showing business performance.
GROUP BY	SQL clause used to group rows sharing a common value.
ORDER BY	SQL clause used to sort query results in ascending or descending order.
HAVING	SQL clause used to filter grouped results based on a condition.
Sub-Category	A finer classification within a main product Category.
Profit Margin %	Ratio of profit to sales, expressed as a percentage.
COGS	Cost of Goods Sold, the direct cost of producing sold products.
Region	A geographic division (e.g., East, West, North, South) used for sales grouping.
Ship Mode	The method of delivery used for an order (e.g., Standard, First Class).
Row ID	A unique identifier assigned to each record/row in the dataset.
Order ID	A unique identifier assigned to each customer order.
Customer ID	A unique identifier assigned to each customer.
Product ID	A unique identifier assigned to each product.
DESC	Descending order, sorting values from highest to lowest.
Dashboard	A visual interface displaying key metrics and data summaries.
RDBMS	Relational database management system

Chapter-1: INTRODUCTION

1.1 INTRODUCTION

In today's competitive business environment, organizations generate vast volumes of transactional data through their day-to-day sales operations. This data, when properly analyzed, holds the key to understanding customer behavior, product performance, regional trends, and overall profitability. The Sales Transaction Analysis project was undertaken with the objective of transforming raw, transaction-level sales data into meaningful, actionable business insights that can support strategic decision-making.

This project focuses on analyzing a comprehensive sales dataset that captures key details such as Order ID, Order Date, Customer, Region, State, Category, Sub-Category, Product Name, Sales, and Profit. By examining these fields collectively, the analysis aims to uncover patterns related to revenue generation, profitability across categories and regions, and the performance of individual products. Such insights are essential for identifying high-performing segments as well as areas that may be contributing to losses, enabling the business to make informed, data-driven decisions.

To carry out this analysis, a structured, multi-tool approach was adopted. MySQL was used to query and aggregate the sales data, allowing for the extraction of key metrics such as profit by sub-category, profit margin percentage by region, the top five most profitable products, profit distribution by category and state, and the identification of loss-making products. Microsoft Excel was utilized to organize, clean, and further explore the dataset, while Power BI was employed to build an interactive dashboard that visually represents these findings through charts, filters, and key performance indicators. Together, these tools provide a complete analytical workflow, from raw data querying to dynamic visualization.

The ultimate goal of this project is to present a clear, comprehensive, and visually engaging analysis of sales performance that highlights profitability trends across different dimensions of the business, namely region, state, category, sub-category, and product. By doing so, this project demonstrates how data analytics can be leveraged to drive better business outcomes, optimize product offerings, and guide regional strategy. The following sections of this report present the detailed findings derived from the SQL analysis, supported by the Excel workbook and the Power BI dashboard.

- The Sales Transaction Analysis project was conducted to transform raw transactional sales data into meaningful business insights for better decision-making.
- The project analyzes sales data fields such as customer details, regions, products, sales, and profit to identify trends and performance patterns.
- MySQL, Microsoft Excel, and Power BI were used collectively for data querying, processing, analysis, and interactive dashboard visualization.
- The main objective of the project is to provide a clear analysis of sales performance and profitability across different business dimensions.

1.2 BACKGROUND STDY

1. Importance of Sales Data Analysis in Retail Business

In today's competitive retail landscape, businesses generate massive volumes of transactional data daily through orders, shipments, and customer interactions. Analyzing this data helps organizations understand purchasing patterns, profitability trends, and operational efficiency. Without structured analysis, valuable insights remain hidden within raw spreadsheets and databases. Companies that leverage data-driven decision-making consistently outperform competitors relying on intuition alone. This project addresses that need by transforming raw sales records into actionable business intelligence.

2. Need for Profit and Margin Analysis

Revenue generation alone does not guarantee business success; profitability is the true indicator of financial health. Many products or regions may show high sales figures but low or negative profit margins due to high costs, discounts, or returns. Understanding profit margin percentage across categories, sub-categories, and products allows management to identify which segments truly contribute to the bottom line. This is especially critical for office supply and furniture retailers where margins can vary significantly. The dataset's inclusion of Cost of Goods and Revenue per Unit columns supports this deeper financial scrutiny.

3. Regional and Geographic Performance Insights

Sales performance often varies significantly across different regions, states, and cities due to demographic, economic, and logistical factors. Identifying high-performing versus underperforming regions enables targeted marketing strategies and resource allocation. The dataset captures Region, State, and City fields, allowing granular geographic segmentation. This regional lens helps businesses decide where to expand operations or where corrective action is needed. Such insights are vital for supply chain optimization and regional inventory planning.

4. Product and Category-Level Performance Evaluation

Not all products contribute equally to a company's success, and some may even generate consistent losses. Analyzing performance at the Category, Sub-Category, and individual Product Name level reveals which items drive growth and which drag down profitability. The SQL queries in this project specifically isolate top-performing and loss-making products. This granular evaluation supports inventory rationalization, pricing adjustments, and discontinuation decisions for underperforming SKUs. It also helps identify star products worth promoting further.

5. Role of Shipping and Logistics in Profitability

Shipping mode and shipping duration directly influence customer satisfaction and operational costs. Faster shipping options like "Same Day" or "First Class" typically cost more, which can erode profit margins if not priced appropriately. The dataset's Ship Mode and Shipping Duration fields allow analysis of how logistics choices impact overall profitability. Understanding this relationship helps businesses balance customer service expectations against cost efficiency. This is an increasingly important consideration as e-commerce demands faster delivery.

6. Customer Segmentation and Behavior Understanding

Customers are typically segmented into groups such as Consumer, Corporate, and Home Office, each exhibiting distinct purchasing behaviors. Understanding which segment generates the most revenue or profit helps tailor marketing and retention strategies. Customer-level data, including unique Customer ID and Name fields, supports deeper behavioral analysis beyond aggregate sales figures. This segmentation also assists in identifying high-value customers for loyalty programs. Such insights ultimately improve customer relationship management and targeted promotions.

7. Application of SQL and Business Intelligence Tools

Modern data analysis relies heavily on SQL for data extraction, aggregation, and filtering before visualization. The accompanying SQL script demonstrates practical queries for profit aggregation, regional filtering, and ranking analysis using GROUP BY, ORDER BY, and HAVING clauses. Pairing SQL-derived insights with Power BI dashboards bridges the gap between raw data processing and interactive visual storytelling. This combination represents an industry-standard workflow used by data analysts worldwide. It demonstrates the practical integration of database querying with business intelligence reporting.

8. Significance of Dashboard-Based Decision Support Systems

Static reports often fail to capture the dynamic, exploratory nature of business decision-making, whereas interactive dashboards empower stakeholders to filter and drill down independently. The Power BI dashboard created in this project consolidates profit, margin, regional, and product-level insights into a single visual interface. This enables managers to make faster, more informed decisions without needing technical SQL expertise. Dashboards also support real-time monitoring of key performance indicators as new data is added. Such decision-support systems are now considered essential tools across retail, e-commerce, and supply chain industries

1.3 NEED AND IMPORTANCE OF DATA ANALYTICS IN SALES

Sales data analytics allows an organization to move beyond simple revenue tracking and answer deeper business questions such as which product categories are most profitable, which regions underperform, which products are sold at a loss, and how shipping and payment preferences affect operational efficiency. Without systematic analysis, such patterns often remain hidden within thousands of individual transaction rows.

The importance of sales transaction analysis can be summarized through the following points:

- It identifies profitable and loss-making product categories, sub-categories and individual products.
- It reveals regional and state-level variation in sales performance and profit margin.
- It supports inventory and pricing decisions by highlighting products that generate high sales volume but low or negative profit.
- It enables monitoring of customer segment behavior (Consumer, Corporate, Home Office) for targeted marketing.
- It provides management with a single, interactive dashboard instead of static, manually prepared reports.

- Reveals which products, categories, or regions are actually profitable vs. just high in sales volume.
- Profit margin % (not raw revenue) guides pricing, promotions, and resource allocation
- Flags loss-making products/regions early before they scale into bigger financial problems
- Highlights top performers (e.g., Top 5 Products by Profit) so effort and budget go where returns are highest
- Dashboards turn raw transactions into instant insights for managers instead of slow manual reporting

These business questions directly motivated the structure of the analysis carried out during the training, and are addressed in detail in Chapters 3 and 4 of this report.

1.4 OBJECTIVES OF THE PROJECT

The semester industrial training was undertaken with the following specific objectives:

- To understand and apply the complete data analytics workflow — data cleaning, exploration, querying and visualization — on a real-world style sales transaction dataset.
- To gain hands-on proficiency in Microsoft Excel for data cleaning, pivot tables and exploratory analysis.
- To learn and apply Structured Query Language (SQL) for aggregating and filtering large transactional datasets.
- To design and build an interactive, multi-page Power BI dashboard incorporating slicers, KPI cards, and multiple chart types.
- To derive actionable business insights regarding profitability by category, sub-category, region, state and product.
- To document the entire process, results and recommendations in a structured technical report.
- To enhance proficiency in data validation and error-checking to ensure accuracy and reliability of reported figures.
- To explore relationships between sales, discount, and profit to understand factors driving profitability.
- To practice using DAX measures and calculated columns within Power BI for deeper analytical insights.
- To develop the ability to filter and segment data dynamically using interactive dashboard slicers and filters.
- To improve data presentation and storytelling skills by converting analytical results into clear business recommendations.
- To gain exposure to real-world business problems such as identifying underperforming products, regions, or categories.
- To build confidence in using analytical tools collaboratively, simulating a real business intelligence reporting environment.
- To understand how data analytics supports strategic decision-making, forecasting and performance monitoring in sales operations.

1.5 SCOPE OF THE PROJECT

The scope of the project was confined to the analysis of the provided Sales Transaction dataset and did not extend to live data integration with any production database. The analysis covered:

- Cleaning and validating 5,901 transaction records across 25 attributes.
- Computation of sales, profit and profit margin at category, sub-category, region, state and product level.
- Identification of top-performing and loss-making products using SQL aggregation queries.
- Design of an interactive Power BI dashboard with drill-down capability by Region, Category and Customer Segment.
- Preparation of business recommendations based on the derived insights.

The scope explicitly excluded predictive modelling (e.g., sales forecasting) and real-time data pipelines, both of which are suggested as directions for future work in Chapter 5.

1.6 TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used in this project to complete data collection, processing, analysis, and visualization effectively.

1.6.1 Microsoft Excel

Microsoft Excel was used in the initial stage of the project for data exploration, cleaning, and validation before deeper analysis. It allowed quick inspection of the raw sales dataset and helped identify inconsistencies early on.

Key Features Used:

- Data viewing and inspection
- Basic statistical analysis
- Data formatting and cleaning
- Table creation and organization

1.6.2 MySQL

MySQL is a relational database management system used to store the cleaned dataset in a structured format. It helps in organizing large datasets and allows efficient data retrieval.

Key Features Used:

- Database creation and management
- Table design and data import
- Efficient query execution and optimization
- Data retrieval and aggregation
- Integration with Python and Power BI

1.6.3 SQL

SQL (Structured Query Language) is used for analyzing the stored data in MySQL. It helps in extracting meaningful insights from large datasets.

Key Features Used:

- Data retrieval using SELECT statements
- Data filtering using WHERE clause
- Grouping and aggregation using GROUP BY
- Sorting data using ORDER BY
- Advanced queries using window functions and CTEs
- Conditional logic using CASE statements

1.6.4 Power BI

Power BI is a business intelligence tool used for creating interactive dashboards and visual reports. It helps in presenting data insights in a visually understandable format.

Key Features Used:

- Data import from MySQL database
- Data transformation and modeling
- Creation of KPI cards and dashboards
- Interactive filters and slicers
- Advanced charts and visual storytelling
- Report publishing and sharing

1.7 OVERVIEW OF THE ANALYSIS

This project analyzed a real-world style sales transaction dataset to understand where profit is actually being generated across the business, and where it's being lost. The analysis moved through six connected stages:

1.Data collection

The raw dataset (Sales_Transaction_Analysis.csv) contained transaction-level sales records including Sales, Profit, Category, Sub-Category, Product Name, Region, and State.

2.Data cleaning and initial exploration (Excel)

Before any deep analysis, the data was reviewed in Excel to check for inconsistencies, missing values, and formatting issues, and to get a quick first look at the data using pivot tables.

3. Structured storage (MySQL)

The cleaned dataset was loaded into a MySQL database, organizing it into a queryable table (sales_transaction_analysis). This made it possible to run repeatable, efficient queries instead of manually filtering in a spreadsheet.

4. Query-based analysis (SQL)

A series of SQL queries were written to answer specific business questions:

- Total profit by Sub-Category and Category — to see which product groupings drive the most profit
- Profit margin % by Region — to compare regional efficiency, not just raw revenue
- Top 5 Products by Profit — to identify the strongest individual performers
- Profit by State — to understand geographic performance
- Most loss-making products (profit < 0) — to flag items losing money despite being sold

5. Key findings from the queries

- Profit is not evenly distributed — some categories and sub-categories contribute disproportionately more to total profit than others
- Regions with similar sales volumes can have very different profit margins, showing that high sales don't always mean high profitability
- A small number of products account for a large share of total profit (the "Top 5"), while a separate, often overlooked, set of products is actively losing money
- State-level profit analysis revealed geographic concentration of both strong and weak performance

6. Visualization (Power BI)

The SQL outputs were connected to an interactive Power BI dashboard with KPI cards, slicers (Region, Category, etc.), and charts, allowing anyone — not just someone who can write SQL — to explore the same insights visually and filter by their area of interest.

Overall conclusion of the analysis: the project shows that sales volume alone is a poor indicator of business health — profit margin, regional efficiency, and individual product performance tell a very different (and more actionable) story, and a structured Excel → SQL → Power BI pipeline is an effective way to surface that story from raw transactional data.

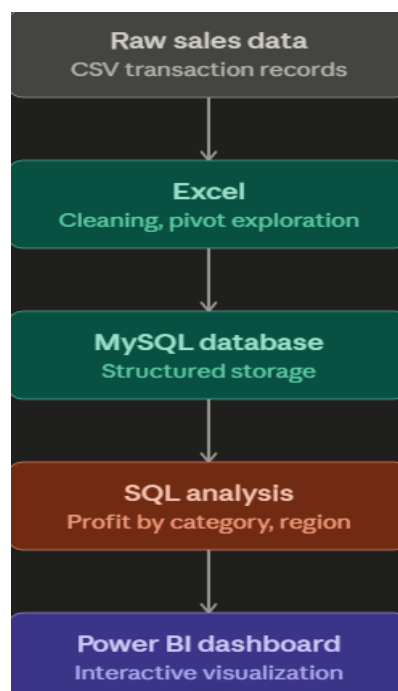


Figure 1.1: Data Analytics Workflow

1.8 PROBLEM STATEMENT

Organizations generate large volumes of sales transaction data every day, but raw transactional records on their own do not reveal where the business is actually making or losing money. Sales teams and management often track total revenue closely, but revenue alone can be misleading — a product, region, or category can show strong sales figures while still generating low or negative profit due to high discounts, costs, or inefficiencies.

In this project, the sales transaction dataset contains detailed records of products sold across multiple categories, sub-categories, regions, and states, along with their corresponding sales and profit values. However, in its raw form, this data is difficult to interpret. It does not clearly show:

- Which categories, sub-categories, or products are genuinely profitable versus which only appear successful based on sales volume
- Which regions and states have strong sales but weak profit margins
- Which specific products are consistently losing money despite being sold regularly
- How profit is distributed unevenly across different segments of the business

Without a structured analysis, these issues can go unnoticed, leading to poor decisions such as continued investment in low-profit or loss-making products, inefficient regional resource allocation, or missed opportunities to scale high-margin products and regions.

The core problem this project addresses is: how can raw, unprocessed sales transaction data be cleaned, structured, and analyzed to accurately identify profit performance across categories, sub-categories, products, regions, and states — and presented in a way that supports clear, fast, data-driven business decisions?

CHAPTER 2: FIELD OF TRAINING

2.1 INTRODUCTION TO FIELD OF TRAINING

Data Analytics is the process of examining, transforming, and interpreting data to discover meaningful information, patterns, and trends that support decision-making. In today's digital world, organizations generate large amounts of data daily. Data analytics helps businesses convert this raw data into valuable insights that improve efficiency, customer satisfaction, and business performance.

This training focused on the complete data analytics process, including data collection, data cleaning, database management, query processing, and dashboard development using industry-standard tools such as Microsoft Excel, Python, MySQL, and Power BI.

2.1.1 Importance of Data Analytics in Modern Business

Data analytics has become essential for businesses across all industries. Organizations use data analytics to:

- Understand customer behavior and preferences
- Optimize operational efficiency
- Improve product and service quality
- Identify market trends and opportunities
- Reduce costs and increase profitability
- Make data-driven strategic decisions

2.1.2 Data Analytics Skills

The training developed the following skills:

- Data cleaning and preprocessing
- Database management and query writing
- Data visualization and dashboard creation
- Business intelligence and reporting
- Problem-solving and analytical thinking

2.2 DATA ANALYTICS AND BUISNESS INTELLIGENCE: OVERVIEW

Data Analytics is the science of examining raw data with the purpose of drawing conclusions about the information it contains. It encompasses several sub-disciplines: descriptive analytics (what happened), diagnostic analytics (why it happened), predictive analytics (what is likely to happen) and prescriptive analytics (what action should be taken). The training undertaken in this project primarily focused on descriptive and diagnostic analytics — summarizing historical sales performance and explaining the drivers of profit and loss across categories, regions and products.

Business Intelligence refers to the technologies, applications and practices used to collect, integrate, analyse and present business data. A BI dashboard is the primary artefact through which BI insights are delivered to decision-makers; it typically combines Key Performance

Indicator (KPI) cards, charts, tables and interactive filters (slicers) on a single screen, allowing users to monitor performance and drill down into specific dimensions without needing to write queries themselves.

The typical analytics workflow followed in industry, and replicated during this training, consists of the following stages:

1. Data collection — obtaining the raw transactional dataset.
2. Data cleaning and preparation — handling missing values, correcting data types, removing duplicates.
3. Exploratory data analysis — using spreadsheet tools to understand distributions and relationships.
4. Structured querying — using SQL to aggregate and filter data to answer specific business questions.
5. Visualisation and dashboard design — using BI tools to present findings interactively.
6. Insight generation and reporting — translating the analysis into business recommendations.

2.3 OVERVIEW OF DATA ANALYTICS

Data Analytics refers to the systematic computational analysis of data to identify useful information and support business decisions. It involves collecting, cleaning, organizing, analyzing, and visualizing data.

2.3.1 Types of Data Analytics

- Descriptive Analytics: Describes what happened in the past
- Diagnostic Analytics: Explains why something happened
- Predictive Analytics: Predicts what will happen in the future
- Prescriptive Analytics: Recommends actions to achieve desired outcomes

2.3.2 Data Analytics Process

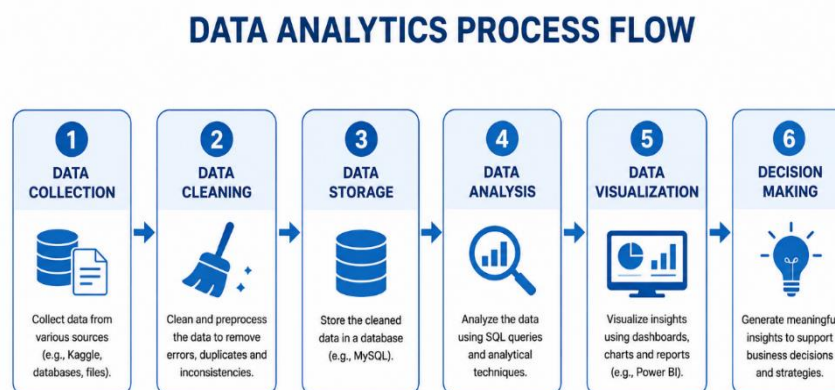


Figure 2.1: Data Analytics Process Flow

2.3.3 Major Stages of Data Analytics

- Data Collection: Gathering data from various sources

- Data Cleaning and Preparation: Removing errors and inconsistencies
- Data Storage and Management: Organizing data for efficient access
- Data Analysis: Applying statistical and computational techniques
- Data Visualization: Presenting results through visual representations

2.4 MICROSOFT EXCEL

Microsoft Excel is one of the most widely used spreadsheet applications for data management and analysis. It provides various tools for organizing, cleaning, and summarizing data.

2.4.1 Key Features Used During Training

- Data Entry and Management
- Creating and formatting worksheets
- Importing and exporting data
- Data validation and error checking
- Data Analysis Tools
- Sorting and filtering data
- Conditional formatting for data visualization
- Pivot Tables for data summarization
- Charts and graphs for visual representation
- Statistical Functions
- SUM, AVERAGE, COUNT, MAX, MIN
- VLOOKUP and HLOOKUP
- IF statements and logical functions
- Statistical analysis tools
- Data Validation
- Setting validation rules
- Creating drop-down lists
- Error checking and data consistency

2.4.2 Role of Excel in the Project

Excel serves as an important foundation for understanding data before moving to advanced analytical tools. In this project, Excel was used for:

- Initial data exploration and understanding
- Basic data validation and quality checks
- Quick summary statistics and analysis

2.5 MySQL (Database Management and Query Processing)

2.5.1 Introduction to MySQL

MySQL is an open-source Relational Database Management System (RDBMS) used for storing, managing, and retrieving structured data efficiently. It uses Structured Query Language

(SQL) to perform various database operations. MySQL is widely used in business applications because of its reliability, scalability, and ease of use.

In this project, MySQL was used to store the cleaned customer shopping behavior dataset and perform analytical queries to extract meaningful insights before creating visualizations in Power BI.

2.5.2 Need for MySQL in the Project

After cleaning the dataset in Excel, the processed data was imported into MySQL. The database acted as a central repository where data could be stored securely and queried efficiently.

MySQL was used for:

- Data Storage and Management
- Data Retrieval and Filtering
- Data Aggregation and Grouping
- Trend Analysis and Reporting
- Supporting Power BI visualization

2.5.3 Installation and Setup

The MySQL Community Server was installed on the local system along with MySQL Workbench.

The setup process involved:

1. Installing MySQL Server.
2. Configuring username and password.
3. Creating a database.
4. Connecting through MySQL Workbench.
5. Importing the cleaned dataset into MySQL.

2.5.4 Advantages of Using MySQL

- Fast query execution.
- Efficient data storage.
- Easy integration with Power BI.
- Supports large datasets.
- Provides secure database management.
- Simplifies business data analysis.

2.5.5 Role of MySQL in the Project

MySQL served as the backbone of the project by storing cleaned data and enabling complex query execution. The results obtained from SQL queries were later connected to Power BI for visualization and dashboard creation.

2.6 POWER BI (Dashboard Creation)

2.6.1 Introduction to Power BI

Power BI is a Business Intelligence and Data Visualization tool developed by Microsoft. It helps users transform raw data into meaningful insights through interactive dashboards, charts, reports, and visualizations.

In this project, Power BI was connected directly to the MySQL database to visualize customer shopping behavior data.

2.6.2 Features Used in the Project

- Data Import from MySQL
- Data Transformation
- Data Modeling
- KPI Cards
- Bar Charts
- Pie Charts
- Line Charts
- Slicers and Filters
- Interactive Dashboard Design

2.6.3 Dashboard Components

The **Sales Transaction Analysis Dashboard** was developed in Microsoft Power BI to provide a comprehensive view of sales and profitability performance. The dashboard consists of two interactive report pages: **Sales Overview Dashboard** and **Profitability Analysis Dashboard**. Each page contains multiple visualizations, KPIs, charts, maps, and slicers that enable users to analyze business performance from different perspectives.

1. Sales Overview Dashboard

The Sales Overview Dashboard focuses on overall sales performance and provides key insights related to revenue generation, order volume, product demand, customer segments, and geographical distribution.

Key Components:

- **Total Sales KPI Card:** Displays the overall sales revenue generated during the selected period.
- **Orders KPI Card:** Shows the total number of orders placed by customers.
- **Profit KPI Card:** Represents the total profit earned from all transactions.
- **Profit Margin KPI Card:** Indicates the percentage of profit earned relative to sales.
- **Quantity KPI Card:** Displays the total quantity of products sold.

Visualizations Included:

- **Sales by Sub-Category (Bar Chart):**
Shows sales contribution of different product sub-categories such as Phones, Chairs, Storage, Accessories, Tables, and Binders. This visualization helps identify top-performing products.
- **Sales Trend (Area/Line Chart):**
Displays monthly sales performance throughout the year, allowing users to identify seasonal patterns and growth trends.
- **Sales by Segment (Donut Chart):**
Represents sales distribution across customer segments such as Consumer, Corporate, and Home Office.
- **Sales by State (Map Visualization):**
Displays geographical sales distribution across different states, helping users identify high-performing regions.

Interactive Filters (Slicers):

- Order Date
- Segment
- Category
- Region

These filters enable dynamic analysis based on user-selected criteria.

2. Profitability Analysis Dashboard

The Profitability Analysis Dashboard focuses on profit generation, profit margins, and identifying products and regions contributing to overall business profitability.

Key Components:

- **Profit KPI Card:** Displays total profit earned.
- **Profit Margin Percentage Card:** Shows overall profit margin.
- **Average Profit per Order Card:** Calculates average profit generated per order.
- **Loss-Making Orders Card:** Displays the number of orders resulting in losses.

Visualizations Included:

- **Profit by Sub-Category (Column Chart):**
Shows profit generated by different product sub-categories. Positive and negative profit contributions can be easily identified.
- **Profit Trend (Area/Line Chart):**
Illustrates monthly profit fluctuations and overall profitability trends over time.
- **Profit Margin by Region (Bar Chart):**
Compares profit margin percentages across different regions such as West, East, South, and Central.
- **Top 5 Products by Profit (Table Visualization):**
Displays the most profitable products along with sales amount, profit earned, and profit margin percentage.

Interactive Filters (Slicers):

- Order Date
- Segment
- Category
- Region

These slicers allow users to perform detailed profitability analysis based on selected business dimensions.

Features of the Dashboard

The dashboard incorporates several advanced Power BI features to enhance usability and decision-making:

- Interactive cross-filtering and cross-highlighting.
- Dynamic KPI calculations using DAX measures.
- Drill-down analysis through charts and visualizations.
- Geographical mapping for regional performance analysis.
- User-friendly navigation between dashboard pages.
- Real-time filtering using slicers.
- Responsive and visually appealing dashboard design.

Business Benefits

The developed dashboard provides significant business value by:

- Monitoring overall sales and profit performance.
- Identifying top-performing and underperforming products.
- Tracking monthly sales and profit trends.
- Evaluating customer segment contributions.
- Analyzing regional performance and profitability.
- Supporting data-driven decision-making processes.
- Improving business intelligence and reporting efficiency.

This dashboard serves as an effective analytical tool for management, sales teams, and decision-makers to monitor business performance and derive actionable insights from transaction data.

2.6.3 Benefits of Power BI

- Easy-to-understand visual reports.
- Interactive data exploration.
- Real-time filtering and analysis.
- Professional dashboard design.
- Faster business decision-making.

Role of Power BI in the Project

Power BI represents the final stage of the data analytics workflow. After data cleaning in Excel and analysis in MySQL, Power BI was used to create interactive dashboards that transformed analytical results into visual insights. These dashboards help users understand transaction analysis.

2.7 DEVELOPMENT TOOLS

The following tools were used during the project development process:

Tool	Used Purpose
Microsoft Excel	Initial Data Analysis and Preparation
MySQL	Database Management and Query Processing
SQL	Data Analysis Queries
Power BI	Dashboard and Report Creation
Kaggle	Dataset Source

Table 1.1: Tools and Technologies Used Purpose

These tools work together to form a complete data analytics workflow.

2.8 LEARNING OUTCOMES

The training provided practical exposure to real-world data analytics techniques. The major learning outcomes include:

- Understanding the complete data analytics lifecycle.
- Learning data cleaning and preprocessing techniques.
- Working with Python libraries for data manipulation.
- Developing SQL query-writing skills.
- Managing databases using MySQL.
- Designing interactive dashboards using Power BI.
- Generating business insights from data.
- Improving problem-solving and analytical thinking abilities.

2.9 ADVANTAGES OF DATA ANALYTICS

Data analytics offers several benefits to organizations:

- Supports data-driven decision-making.
- Improves business performance.
- Identifies customer preferences and behavior.
- Increases operational efficiency.
- Detects trends and patterns.
- Reduces business risks.
- Enhances customer satisfaction.
- Helps predict future outcomes through historical data analysis.

CHAPTER 3: TRAINING WORK UNDERTAKEN

This chapter describes, in sequence, the methodology followed during the training: understanding the dataset, cleaning and preparing the data, writing and validating SQL queries, and designing the Power BI dashboard.

3.1 INTRODUCTION

During the training period, a Sales Transaction Analysis project was developed to analyze business performance using sales data. The project involved collecting transaction records, processing and organizing the data, performing SQL-based analysis, and creating visual dashboards using Power BI.

The objective of this project was to extract meaningful insights from sales transactions such as total sales, profit trends, customer segments, product performance, and regional performance. SQL queries were used for data extraction and analysis, while Power BI was used to visualize the results.

3.1.1 PROJECT WORK FLOW

1. **Data Collection**
 - Collected sales transaction data from Excel/CSV dataset
 - Gathered information such as products, customers, sales, profit, and regions
2. **Data Cleaning & Preprocessing**
 - Removed duplicate records
 - Handled missing values
 - Corrected inconsistent data entries
 - Standardized data formats and column names
 - Validated data quality
3. **Database Management**
 - Created database tables in SQL
 - Imported cleaned data into database
 - Organized and stored records efficiently
 - Managed data using SQL queries
4. **SQL Analysis**
 - Calculated total sales and profit
 - Performed category-wise analysis
 - Conducted region-wise analysis
 - Identified top-performing products
 - Generated customer and sales insights
5. **Data Visualization**
 - Created Power BI dashboards
 - Developed KPI cards and charts
 - Added filters and slicers
 - Built interactive reports
6. **Business Insights & Decision Making**
 - Identified sales trends
 - Analyzed customer behavior

- Determined best-selling products
- Supported business decision-making

3.2 DATA COLLECTION AND CLEANING MODULE

The Data Collection and Cleaning Module was responsible for gathering and preparing the sales transaction dataset for further analysis.

The dataset included various attributes such as:

- Order Date
- Product Name
- Category
- Sub-Category
- Segment
- Region
- State
- Sales
- Profit

Several data preprocessing techniques were applied before analysis:

- Removing duplicate records
- Handling missing values
- Correcting inconsistent entries
- Standardizing column formats
- Validating data quality
- Converting date fields into appropriate date formats

These cleaning procedures improved the reliability and accuracy of the data before loading it into the database.

3.3 DATASET DESCRIPTION

The dataset used for this project, Sales Transaction Analysis, contains 5,901 transaction-level records and 25 attributes, spanning the period from 1 January 2019 to 31 December 2020 and representing 3,003 unique customer orders. The table below describes each field in the dataset.

Field	Description
Row ID / Order ID	Unique identifiers for the line item and the parent order.
Order Date / Ship Date	Date on which the order was placed and shipped.
Ship Mode	Shipping service level: Standard Class, Second Class, First Class, Same Day.
Customer ID / Name / Segment	Customer identity and business segment: Consumer, Corporate, Home Office.
Country / City / State / Region	Geographic location of the order.

Product ID / Category / Sub-Category / Product Name	Product hierarchy: Furniture, Office Supplies, Technology.
Sales / Quantity / Profit	Transaction sales value, units sold, and profit earned.
Profit Margin %	Profit expressed as a percentage of Sales.
Returns	Flag indicating whether the item was returned.
Payment Mode	Mode of payment: Cards, Online, COD.
Shipping Duration (Days)	Number of days between order date and ship date.
Cost of Goods / Revenue per Unit	Derived cost and per-unit revenue figures.

Table 3.1: Dataset Field Description

3.4 DATA CLEANING AND PREPRATION IN EXCEL

Before any analysis could be carried out, the raw dataset was cleaned and validated in Microsoft Excel. The following steps were carried out by the trainee:

1. Verified that Order Date and Ship Date columns were stored as proper date values, and recalculated Shipping Duration where inconsistent.
2. Checked the Sales, Profit, Quantity and Cost of Goods columns for non-numeric or blank entries and corrected formatting issues.
3. Standardized categorical text fields (Region, Category, Sub-Category, Segment, Payment Mode, Ship Mode) to ensure consistent spelling and capitalisation, which is essential before any GROUP BY aggregation in SQL.
4. Treated the Returns column, which contained a flag for returned items and blank values for non-returned items, ensuring blanks were correctly interpreted as “not returned” rather than missing data. Of the 5,901 records, 287 were flagged as returns.
5. Created calculated columns where required, including Profit Margin % ($\text{Profit} / \text{Sales} \times 100$) and verified the existing Cost of Goods and Revenue per Unit columns against manual precomputation for a sample of rows.
6. Built PivotTables to obtain a first-pass summary of Sales and Profit by Category and Region, which was later used to cross-validate the SQL query results in Section 3.3.

This cleaning stage ensured that the dataset loaded into the SQL database in the next stage was consistent and ready for accurate aggregation. A screenshot of the cleaned worksheet, showing the standardized columns and the Profit Margin % calculated column, is presented in Figure 3.1.

Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Profit	Returns	Profit Margin %	Payment
4696	OFF-BI-10002309	Office Supply Binders	Avery Heavy-Duty EZD Binder With Locking Rings	22.32	4	10.7136			48 Cards
4697	OFF-LA-10003923	Office Supply Labels	Alphabetical Labels for Top Tab Filing	103.6	7	51.8			50 Cards
4698	TEC-AC-10003433	Technology Accessories	Maxell 4.7GB DVD+R 5/Pack	85.97	3	1.3365			1.554612074 Online
4699	TEC-PH-10001363	Technology Phones	Apple iPhone 5S	606.99	1	170.997			28.1713043 COD
4700	FUR-FU-10000820	Furniture Furnishings	Tensor Brushed Steel Torchieri Floor Lamp	139.97	3	9.1746			6.554690291 Cards
4701	FUR-TA-10004767	Furniture Tables	Safco Drafting Table	337.116	6	-4.2588			-1.263304026 Cards
4702	OFF-AR-10001955	Office Supply Art	Newell 319	59.52	3	17.856			30 Online
4703	FUR-CH-10004886	Furniture Chairs	Bevis Steel Folding Chairs	230.28	3	23.028			10 COD
4704	OFF-PA-10003641	Office Supply Paper	Xerox 1909	105.52	5	34.294			32.5 COD
4705	OFF-EN-10003134	Office Supply Envelopes	Staple envelope	115.36	2	11.68	1		10.12482663 Online
4706	TEC-PH-10002564	Technology Phones	OtterBox Defender Series Case - Samsung Galaxy S4	161.976	3	8.997	1		5.554526597 Online
4707	OFF-PA-10003893	Office Supply Paper	Xerox 1962	119.56	2	3.852	1		3.221813315 COD
4708	OFF-BI-10000174	Office Supply Binders	Wilson Jones Clip & Carry Folder Binder Tool for Ring Binders, Clear	104.92	3	4.872	1		4.643537934 Cards
4709	OFF-AP-10001058	Office Supply Appliances	Sanyo 2.5 Cubic Foot Mid-Size Office Refrigerators	2547.29	9	654.7554	1		25.70399915 COD
4710	OFF-ST-10001496	Office Supply Storage	Standard Rollaway File with Lock	568.57	3	140.5482	1		24.71959477 COD
4711	OFF-BI-10004230	Office Supply Binders	GBC Recycled Grain Textured Covers	264.056	8	77.3696	1		29.30045142 Cards
4712	FUR-CH-10001146	Furniture Chairs	Global Task Chair, Black	81.424	2	-9.1602			-11.25 COD
4713	OFF-ST-10001590	Office Supply Storage	Tenex Personal Project File with Scoop Front Design, Black	134.8	10	35.048			26 Online
4714	OFF-BI-10003684	Office Supply Binders	Wilson Jones Legal Size Ring Binders	418.582	9	-59.373	1		-14.18431753 Online
4715	OFF-SU-10000946	Office Supply Supplies	Staple remover	433.688	7	5.0274	1		1.159220453 Online
4716	OFF-AR-10001955	Office Supply Art	Newell 319	59.52	3	17.856			30 Online

Figure 3.1: Cleaned Sales Transaction dataset in Microsoft Excel

To prepare the dataset for the Power BI dashboard, the same cleaned data was imported into Power Query Editor, where additional transformation steps — promoting headers, setting data types, removing duplicate rows, handling blank Returns values, and adding the Profit Margin % custom column — were applied and recorded as a repeatable sequence of Applied Steps, as shown in Figure 3.2.

Row ID	Order ID	Order Date	Ship Date	Ship Mode
1	4918	CA-2019-160304	1/1/2019	1/7/2019 Standard Class
2	4919	CA-2019-160304	1/2/2019	1/7/2019 Standard Class
3	4920	CA-2019-160304	1/2/2019	1/7/2019 Standard Class
4	3074	CA-2019-125206	1/3/2019	1/5/2019 First Class
5	8604	US-2019-116365	1/3/2019	1/8/2019 Standard Class
6	8605	US-2019-116365	1/3/2019	1/8/2019 Standard Class
7	8606	US-2019-116365	1/3/2019	1/8/2019 Standard Class
8	9494	CA-2019-105207	1/3/2019	1/8/2019 Standard Class
9	9495	CA-2019-105207	1/3/2019	1/8/2019 Standard Class
10	2898	US-2019-164630	1/4/2019	1/9/2019 Standard Class
11	5868	CA-2019-158211	1/4/2019	1/8/2019 Standard Class
12	5869	CA-2019-158211	1/4/2019	1/8/2019 Standard Class
13	863	CA-2019-134474	1/5/2019	1/7/2019 Second Class
14	864	CA-2019-134474	1/5/2019	1/7/2019 Second Class
15	865	CA-2019-134474	1/5/2019	1/7/2019 Second Class
16	2162	CA-2019-101938	1/7/2019	1/12/2019 Standard Class
17	8031	CA-2019-158806	1/7/2019	1/11/2019 Standard Class
18	8032	CA-2019-158806	1/7/2019	1/11/2019 Standard Class
19	6851	US-2019-100461	1/8/2019	1/12/2019 Standard Class
20	6852	US-2019-100461	1/8/2019	1/12/2019 Standard Class
21	7808	US-2019-137295	1/8/2019	1/13/2019 Standard Class

Figure 3.2: Power Query Editor showing the Applied Steps used to clean and transform the dataset before loading into the Power BI data model

3.5 SQL-BASED ANALYSIS

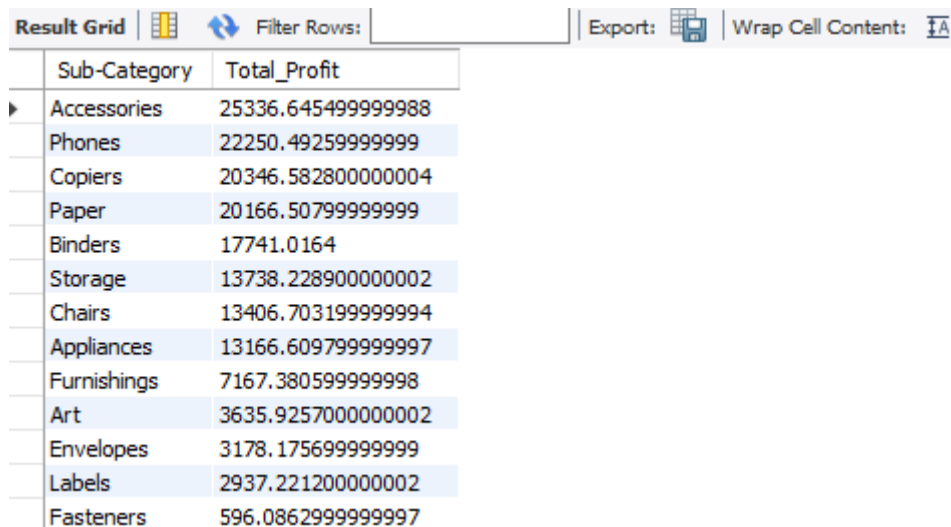
The cleaned dataset was loaded into a MySQL schema named sales_trasaction_analysics, into a table named data. A set of seven SQL queries was then designed and executed to answer the core business questions of the project. Each query is described below, with the full SQL code reproduced in Appendix A.

3.3.1 Query 1: Profit by Sub-Category

This query grouped all transactions by the Sub-Category field and summed the Profit column, ordering the results in descending order of total profit. This identified which product sub-categories (e.g., Accessories, Phones, Tables) were the strongest and weakest contributors to overall profitability.

QUERY:

```
SELECT
    `Sub-Category`, SUM(Profit) AS Total_Profit
FROM
    connect.sales_trasaction_analysis
GROUP BY `Sub-Category`
ORDER BY Total_Profit DESC;
```



Sub-Category	Total_Profit
Accessories	25336.645499999988
Phones	22250.492599999999
Copiers	20346.582800000004
Paper	20166.507999999999
Binders	17741.0164
Storage	13738.228900000002
Chairs	13406.703199999994
Appliances	13166.609799999997
Furnishings	7167.380599999998
Art	3635.9257000000002
Envelopes	3178.1756999999999
Labels	2937.2212000000002
Fasteners	596.0862999999997

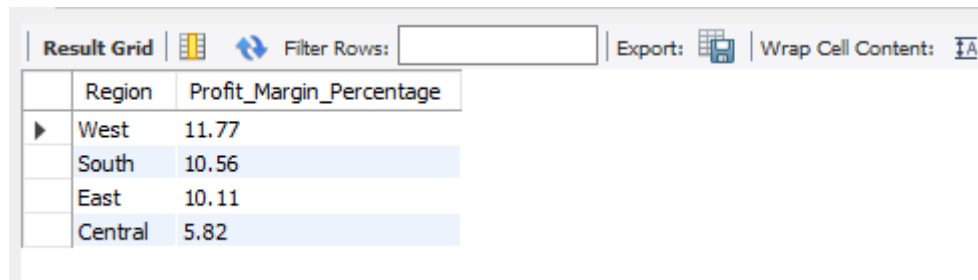
Figure 3.3: MySQL Workbench output for Query 1 — Profit by Sub-Category

3.3.2 Query 2: Profit Margin % by Region

This query computed the overall profit margin percentage for each of the four regions (South, West, Central, East) by dividing the sum of Profit by the sum of Sales for each region, then multiplying by 100 and rounding to two decimal places.

QUERY:


```
SELECT
    Region,
    ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Profit_Margin_Percentage
FROM
    connect.sales_transaction_analysis
GROUP BY Region
ORDER BY Profit_Margin_Percentage DESC;
```



Region	Profit_Margin_Percentage
West	11.77
South	10.56
East	10.11
Central	5.82

Figure 3.4: MySQL Workbench output for Query 2 — Profit Margin % by Region

3.3.3 Query 3: Top 5 Products by Profit

This query grouped transactions by Product Name, computed total Sales and total Profit for each product, derived the profit margin percentage, and used ORDER BY combined with LIMIT 5 to shortlist the five most profitable products in the dataset.

QUERY:

```
SELECT
    `Product Name`,
    SUM(Sales) AS Total_Sales,
    SUM(Profit) AS Total_Profit,
    ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Profit_Margin_Percentage
FROM
    connect.sales_transaction_analysis
GROUP BY `Product Name`
ORDER BY Total_Profit DESC
LIMIT 5;
```

Product Name	Total_Sales	Total_Profit	Profit_Margin_Percentage
Hewlett Packard LaserJet 3310 Copier	13837.732	6407.8932	46.31
3D Systems Cube Printer, 2nd Generation, Mag...	14334.89	3717.9714000000004	25.94
HP Designjet T520 Inkjet Large Format Printer -...	4749.95	2799.984	58.95
Canon imageCLASS 2200 Advanced Copier	14076.824	2771.928	19.69
GBC DocuBind TL300 Electric Binding System	12890.258	2753.7592999999997	21.36

Figure 3.5: MySQL Workbench output for Query 3 — Top 5 Products by Profit

3.3.4 Query 4: Region Filter (Distinct Regions)

This query used SELECT DISTINCT to retrieve the unique list of Region values present in the dataset. This list was used to configure the Region slicer in the Power BI dashboard, ensuring the filter options matched the actual data exactly.

QUERY:

```
SELECT DISTINCT
```

```
    Region
```

```
FROM
```

```
    connect.sales_trasaction_analysis
```

```
ORDER BY Region;
```

Region
Central
East
South
West

Figure 3.6: MySQL Workbench output for Query 4 — Region Filter

3.3.5 Query 5: Profit by Category

This query aggregated Profit at the higher Category level (Furniture, Office Supplies, Technology), providing a top-level summary that complements the more granular Sub-Category analysis in Query 1.

QUERY:

```
SELECT
```

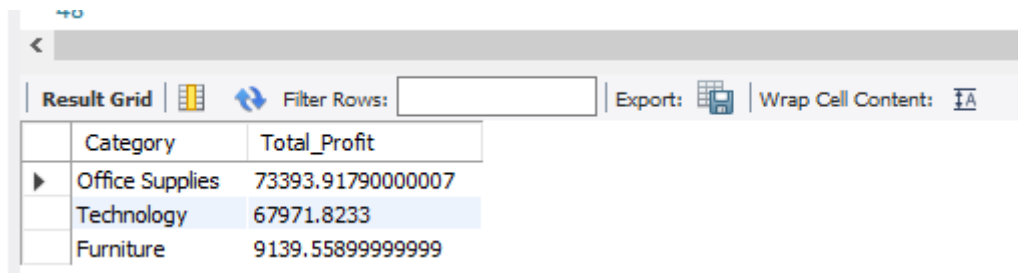
```
    Category, SUM(Profit) AS Total_Profit
```

```
FROM
```

```
    connect.sales_trasaction_analysis
```

```
GROUP BY Category
```

ORDER BY Total_Profit DESC;



Category	Total_Profit
Office Supplies	73393.91790000007
Technology	67971.8233
Furniture	9139.558999999999

Figure 3.7: MySQL Workbench output for Query 5 — Profit by Category

3.3.6 Query 6: Profit by State

This query aggregated Profit by State, allowing identification of the best- and worst-performing states. This was used to construct the state-level profitability ranking discussed in Section 4.6.

QUERY:

SELECT

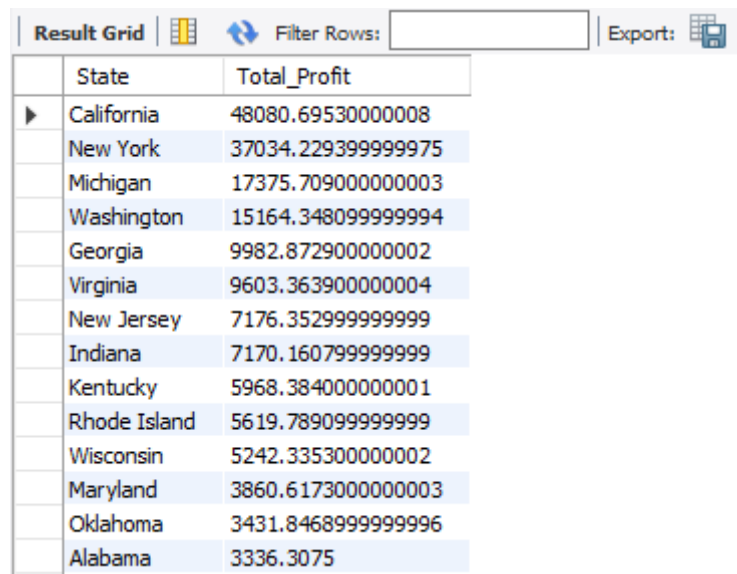
State, SUM(Profit) AS Total_Profit

FROM

connect.sales_trasaction_analysis

GROUP BY State

ORDER BY Total_Profit DESC;



State	Total_Profit
California	48080.695300000008
New York	37034.229399999975
Michigan	17375.709000000003
Washington	15164.348099999994
Georgia	9982.872900000002
Virginia	9603.363900000004
New Jersey	7176.352999999999
Indiana	7170.160799999999
Kentucky	5968.384000000001
Rhode Island	5619.789099999999
Wisconsin	5242.335300000002
Maryland	3860.6173000000003
Oklahoma	3431.8468999999996
Alabama	3336.3075

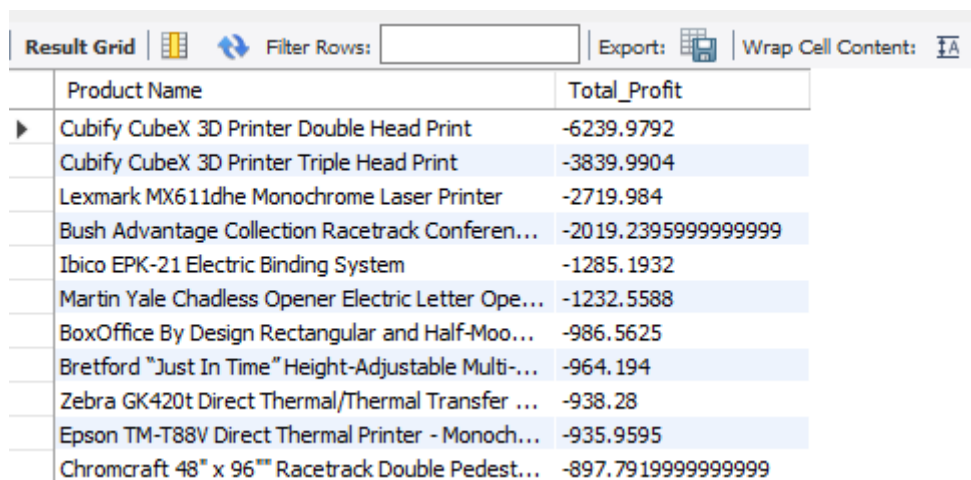
Figure 3.8: MySQL Workbench output for Query 6 — Profit by States

3.3.7 Query 7: Most Loss-Making Products

This query grouped transactions by Product Name and used the HAVING clause to filter only those products whose total summed Profit was negative, ordering the result so that the most severe loss-making products appeared first. This directly supports the business recommendation to review pricing or discontinue these products, discussed in Chapter 5.

QUERY:

```
SELECT
    `Product Name`, SUM(Profit) AS Total_Profit
FROM
    connect.sales_trasaction_analysis
GROUP BY `Product Name`
HAVING SUM(Profit) < 0
ORDER BY Total_Profit;
```



Product Name	Total_Profit
Cubify CubeX 3D Printer Double Head Print	-6239.9792
Cubify CubeX 3D Printer Triple Head Print	-3839.9904
Lexmark MX611dhe Monochrome Laser Printer	-2719.984
Bush Advantage Collection Racetrack Conferen...	-2019.2395999999999
Ibico EPK-21 Electric Binding System	-1285.1932
Martin Yale Chadless Opener Electric Letter Ope...	-1232.5588
BoxOffice By Design Rectangular and Half-Moo...	-986.5625
Bretford "Just In Time" Height-Adjustable Multi-...	-964.194
Zebra GK420t Direct Thermal/Thermal Transfer ...	-938.28
Epson TM-T88V Direct Thermal Printer - Monoch...	-935.9595
Chromcraft 48" x 96" Racetrack Double Pedest...	-897.7919999999999

Figure 3.9: MySQL Workbench output for Query 7 — Most Loss-Making Products (top rows)

3.6 POWER BI DASHBOARD DESIGN

Following the SQL-based analysis, the cleaned dataset was imported into Microsoft Power BI to design an interactive dashboard. The dashboard was developed to convert raw analytical outputs into meaningful visual insights that support business decision-making. Multiple report pages were created to organize the analysis effectively and improve user navigation.

The Power BI dashboard creation process involved the following activities:

- Importing the cleaned dataset from Excel into Power BI.
- Verifying and managing data relationships between tables.
- Creating calculated measures and KPI metrics using DAX functions.
- Developing KPI cards for Total Sales, Orders, Profit, Profit Margin %, Quantity Sold, Average Profit per Order, and Loss-Making Orders.
- Designing interactive visualizations including bar charts, line charts, donut charts, maps, and tables.
- Creating filters and slicers for dynamic data exploration.
- Applying themes, colors, formatting, and dashboard layouts for better visual appearance.

- Organizing visuals across multiple report pages for easier analysis.
- Enabling cross-filtering and drill-down functionality for interactive analysis.
- Testing dashboard responsiveness and validating results against SQL query outputs.

The dashboard was structured across multiple report pages to keep the analysis organized and easy to navigate:

3.4.1 Chart Descriptions

1. KPI Cards

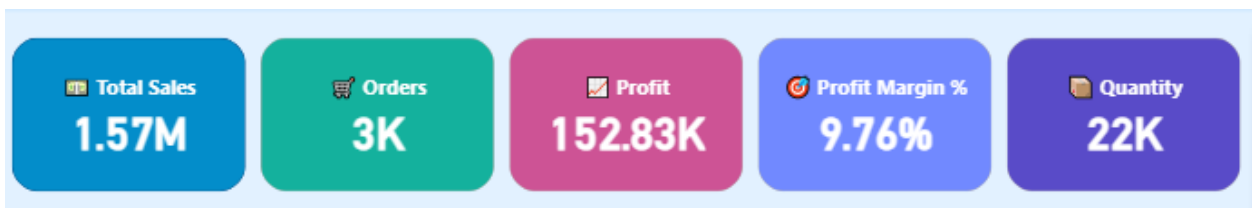


Figure 3.10: KPI Cards

- Display important business metrics.
- Show Total Sales, Orders, Profit, Profit Margin %, and Quantity Sold.
- Provide a quick overview of business performance.

2. Sales by Sub-Category Chart

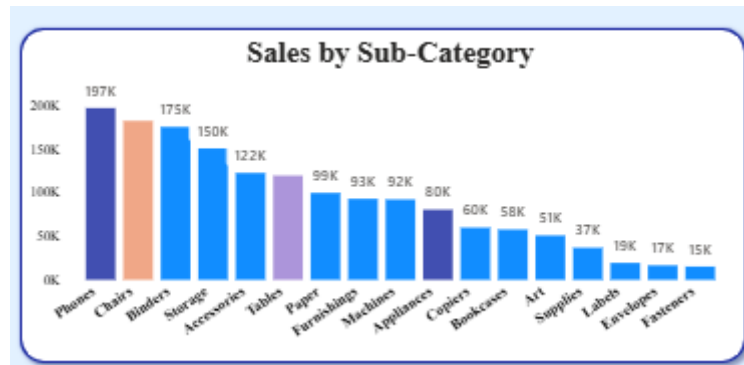


Figure 3.11: Sales by sub-Category

- Displays sales values for different product sub-categories.
- Helps identify top-selling and low-selling products.
- Supports product performance analysis.

3. Sales Trend Chart



Figure 3.12: Sales Trend Chart

- Shows monthly sales performance over time.
- Helps identify growth patterns and seasonal trends.
- Supports future sales forecasting.

4. Sales by Segment Chart

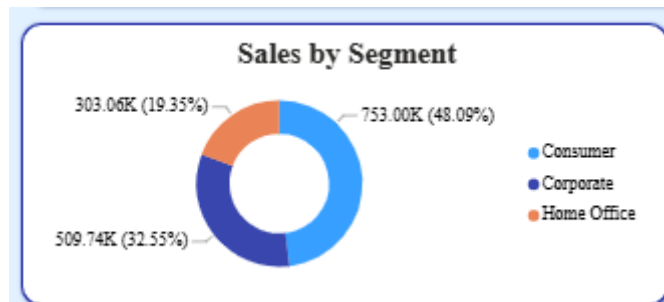


Figure 3.13: Sales by Segment

- Represents sales contribution by customer segments.
- Includes Consumer, Corporate, and Home Office categories.
- Helps understand customer purchasing behavior.

5. Sales by State Map



Figure 3.14: Sales by State

- Displays geographical sales distribution across states.
- Helps identify high-performing and low-performing regions.
- Supports regional business analysis.

6. Profit by Sub-Category Chart

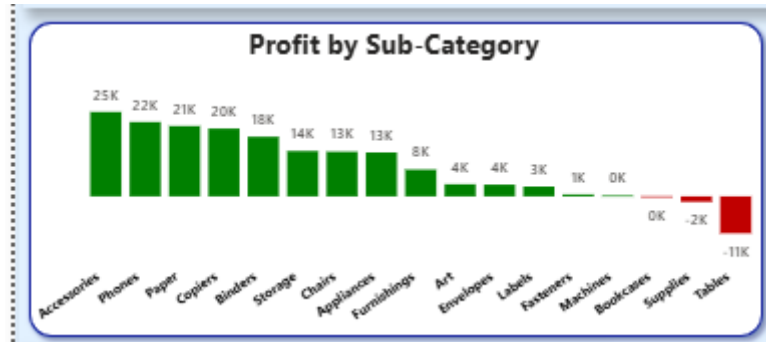


Figure: 3.15: Profit by Sub-Category Chart

- Shows profit generated by each product sub-category.
- Identifies profitable and loss-making products.
- Helps improve product strategies.

7. Profit Trend Chart



Figure 3.16: Profit Trend

- Displays monthly profit changes over time.
- Helps monitor profit growth patterns.
- Supports business performance evaluation.

8. Profit Margin by Region Chart

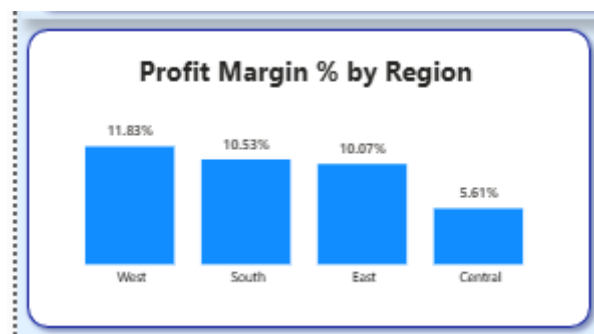


Figure 3.17: Profit Margin % by Region

- Compares profit margins across different regions.

- Identifies regions with better profitability.
- Supports regional decision-making.

9. Top 5 Products by Profit Table

Top 5 Products by Profit			
Product Name	Sales	Profit	Profit Margin %
3D Systems Cube Printer, 2nd Generation, Magenta	14,334.89	3,717.97	25.94%
Canon imageCLASS 2200 Advanced Copier	14,076.82	2,771.93	19.69%
GBC DocuBind TL300 Electric Binding System	12,890.26	2,753.76	21.36%
Hewlett Packard LaserJet 3310 Copier	13,837.73	6,407.89	46.31%
HP Designjet T520 Inkjet Large Format Printer - 24" Color	4,749.95	2,799.98	58.95%
Total	59,889.65	18,451.54	30.81%

Figure 3.18: Top 5 Products by Profit

- Displays products generating maximum profit.
- Includes sales and profit information.
- Helps identify best-performing products.

10. Slicers and Filters

The image shows a dashboard filter and slicer interface. It is organized into several sections:

- Order Date:** Contains three sub-sections:
 - Year:** A dropdown menu currently showing '2019'.
 - Quarter:** A dropdown menu currently showing 'Qtr 1'.
 - Month:** Two dropdown menus showing 'January' and 'February'.
- Segment:** A single dropdown menu currently showing 'All'.
- Category:** Three stacked buttons labeled 'Furniture', 'Office Supplies', and 'Technology'.
- Region:** Four buttons arranged in a 2x2 grid, labeled 'Central', 'East', 'South', and 'West'.

Figure 3.19: Dashboard Filters and Slicers

- Used for dynamic dashboard interaction.
- Includes Order Date, Category, Segment, and Region filters.
- Automatically update all charts and KPI values dynamically.

Interactivity was built into the dashboard through slicers for Region, Category, Segment, and Order Date, allowing all charts and KPI cards to update automatically according to user selections.

3.4.2 Sales Overview Page

This page provides an overall business performance summary and includes:

- KPI cards displaying Total Sales, Orders, Profit, Profit Margin %, and Quantity Sold.
- Sales by Sub-Category bar chart.
- Monthly Sales Trend chart.
- Sales by Segment donut chart.
- Sales by State geographical map.
- Interactive filters for Order Date, Segment, Category, and Region.

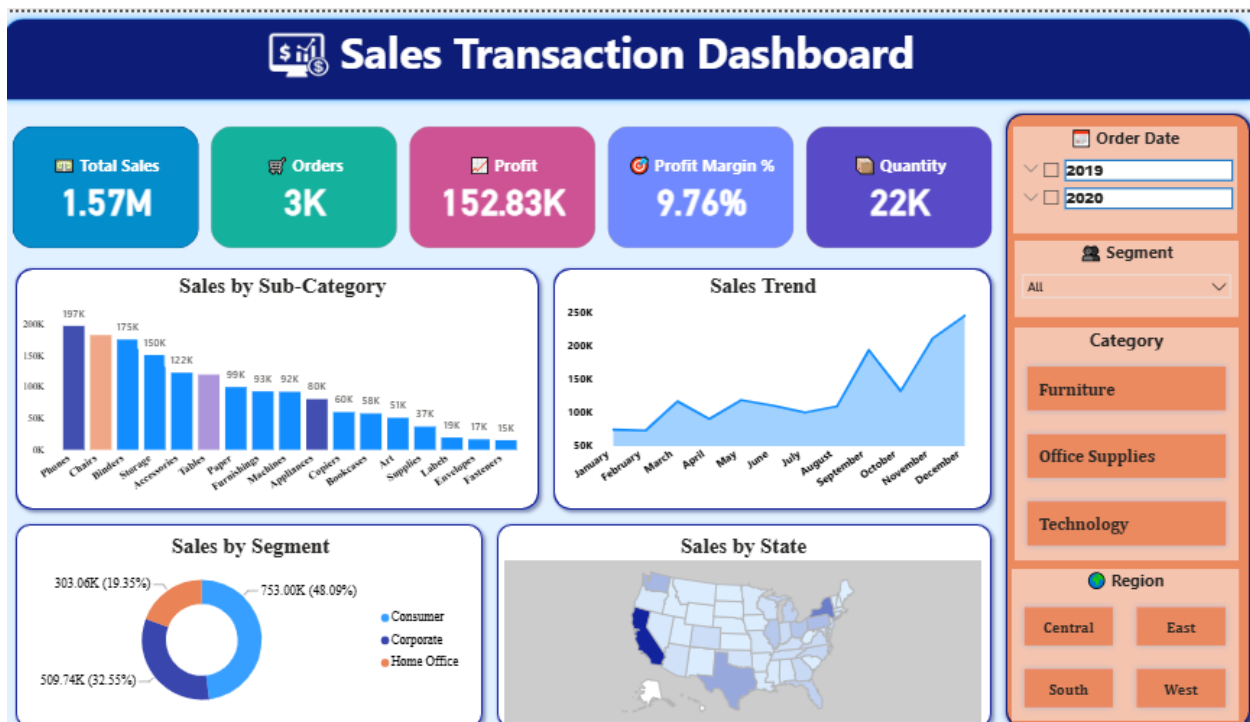


Figure 3.20: Sales Overview Dashboard

3.4.1 Profitability Analysis Page

This page focuses on profitability and performance analysis and includes:

- KPI cards showing Profit, Profit Margin %, Average Profit per Order, and Loss-Making Orders.
- Profit by Sub-Category chart.
- Monthly Profit Trend chart.
- Profit Margin Percentage by Region chart.
- Top 5 Products by Profit table.
- Interactive slicers for dynamic filtering.

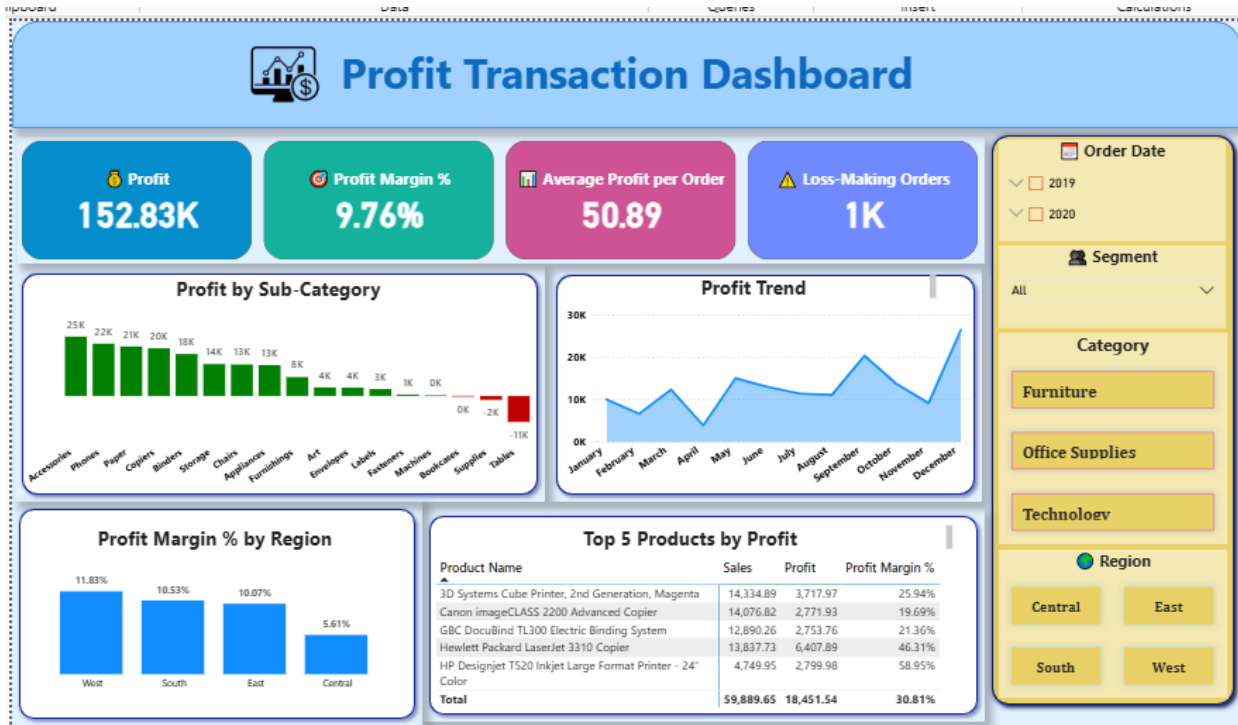


Figure 3.21: Profitability Analysis Dashboard

Interactivity was built into the dashboard through slicers for Region, Category, Segment, and Order Date, allowing all charts and KPI cards to update dynamically based on user selections.

This design approach enables different stakeholders to use the same dashboard according to their requirements. For example, regional managers can filter results for specific regions, while product managers can analyze individual categories without requiring separate reports.

The Power BI dashboard transformed static SQL outputs into an interactive and user-friendly analytical platform that improved accessibility, understanding, and business decision-making.

Following the SQL-based analysis, the cleaned dataset was imported into Microsoft Power BI to design an interactive dashboard. The dashboard was structured across multiple report pages to keep the analysis organized and easy to navigate:

1. Overview page — KPI cards for Total Sales, Total Profit, Overall Profit Margin %, Total Orders and Average Shipping Duration, accompanied by a monthly sales and profit trend line chart.
2. Category and Sub-Category page — bar charts of Sales and Profit by Category and Sub-Category, replicating and visualizing the results of SQL Queries 1 and 5.
3. Regional analysis page — a map/bar chart of Sales by Region together with the Profit Margin % by Region chart (Query 2), and a Region slicer for interactive filtering.
4. Product analysis page — Top 5 most profitable products (Query 3) and a table of loss-making products (Query 7).
5. Customer and Operations page — Sales and Profit by Segment, distribution of orders by Ship Mode and Payment Mode, and a Returns summary.

Interactivity was built into the dashboard through slicers for Region, Category and Segment, allowing any KPI card or chart on the page to be filtered dynamically. This design choice ensures that the same dashboard can serve multiple stakeholders — for example, a regional manager can filter to their own region, while a category manager can filter to their product category — without requiring separate static reports.

3.7 ROLE OF TRAINEE

Throughout the training period, the trainee was actively involved in the complete data analytics workflow and was responsible for executing each stage of the project from data preparation to dashboard development. The trainee independently performed data cleaning, SQL analysis, visualization development, and documentation activities.

The responsibilities undertaken during the training period included:

- Collecting and understanding the sales transaction dataset and its structure.
- Cleaning and validating raw data using Microsoft Excel.
- Identifying and removing duplicate records from the dataset.
- Handling missing values and correcting inconsistent data entries.
- Formatting and standardizing data fields for consistency.
- Verifying data accuracy before further processing and analysis.
- Importing cleaned data into the SQL database environment.
- Creating and managing database tables.
- Writing SQL queries for extracting and analyzing business information.
- Performing category-wise, region-wise, product-wise, and customer-wise analysis.
- Calculating key business metrics such as Total Sales, Profit, and Profit Margin Percentage.
- Cross-validating SQL query outputs using Excel Pivot Tables to ensure accuracy and consistency of results.
- Importing the processed dataset into Microsoft Power BI.
- Designing and developing multiple dashboard pages for analysis.
- Creating KPI cards, charts, tables, maps, and other visual components.
- Developing DAX measures and calculated fields for dashboard metrics.
- Implementing slicers and filters for interactive data analysis.
- Applying dashboard themes, formatting, and layout optimization.
- Testing dashboard functionality and validating analytical outputs.
- Interpreting results and generating meaningful business insights.
- Preparing technical documentation and maintaining records of all activities performed during training.
- Compiling the project report and documenting each stage of the implementation process.

3.8 DATA FLOW DIAGRAMS

To clarify how the raw transaction data was transformed into the final business insights presented in this report, two data flow diagrams were prepared. These diagrams summaries the movement of data across the three tools used during the training — Microsoft Excel, MySQL and Microsoft Power BI — and show how individual data attributes were progressively cleaned, aggregated and visualized.

The data flow diagrams summarize the interaction between Microsoft Excel, MySQL, and Microsoft Power BI and provide a structured view of the complete analytics workflow.

The following stages are represented in the data flow process:

1. Data Collection Stage

- Raw sales transaction data collected from the source dataset.
- Dataset containing information such as Order Date, Product, Category, Sub-Category, Segment, Region, Sales, Profit, Payment Mode, and Ship Mode.
- Initial verification of dataset structure and attributes.

2. Data Cleaning and Preparation Stage (Microsoft Excel)

- Removal of duplicate records.
- Identification and handling of missing values.
- Correction of inconsistent data entries.
- Standardization of column names and formats.
- Validation of date and numerical fields.
- Verification of categorical fields such as Region, Category, Segment, and Payment Mode.
- Creation of calculated columns such as Profit Margin Percentage.

3. Database Management and SQL Processing Stage (MySQL)

- Importing cleaned data into MySQL database tables.
- Organizing and storing transaction records.
- Executing SQL queries for data analysis.
- Performing aggregation and filtering operations.
- Calculating Total Sales, Profit, and Profit Margin values.
- Conducting region-wise, category-wise, customer-wise, and product-wise analysis.

4. Power BI Dashboard Development Stage

- Importing processed data into Power BI.
- Creating relationships and data models.
- Developing DAX measures and calculated metrics.
- Creating KPI cards and interactive visualizations.
- Designing charts, tables, maps, and dashboard layouts.
- Implementing filters and slicers for interactive analysis.

5. Business Insights and Reporting Stage

- Identification of sales trends and patterns.
- Analysis of customer purchasing behavior.
- Identification of profitable and loss-making products.
- Evaluation of regional business performance.
- Supporting business decision-making through interactive dashboards.

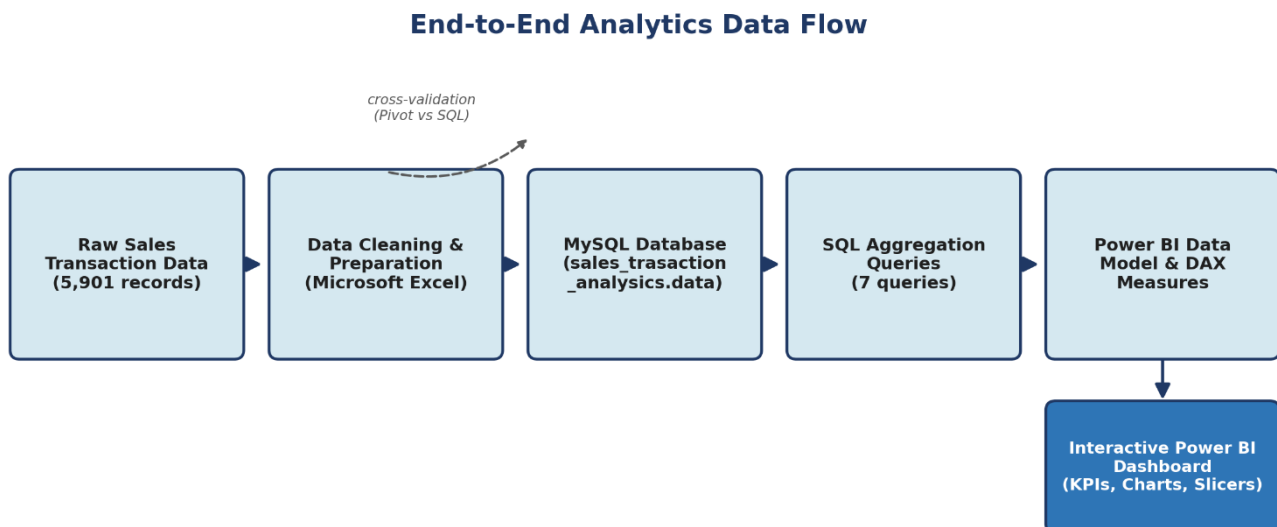


Figure 3.22: End-to-End Analytics Data Flow from Raw Transaction Data through Excel Cleaning, MySQL Database Processing, SQL Analysis, and Power BI Dashboard Visualization

As shown in Figure 3.7, raw transaction records first pass through the cleaning and preparation stage in Microsoft Excel, where important fields such as dates, numerical values, and categorical attributes are validated and standardized. The cleaned dataset is then imported into the MySQL database environment, where SQL queries perform aggregation, filtering, and analytical operations required to answer the project objectives.

The outputs generated through SQL analysis are imported into Power BI for dashboard development. Within Power BI, DAX measures and visual components are developed to convert analytical results into interactive visual representations. Additionally, Excel Pivot Tables were used to cross-check SQL outputs and ensure consistency and accuracy across the analytical process.

The data flow diagrams provide a clear understanding of how raw data is transformed into actionable business insights through a systematic and structured analytics pipeline.

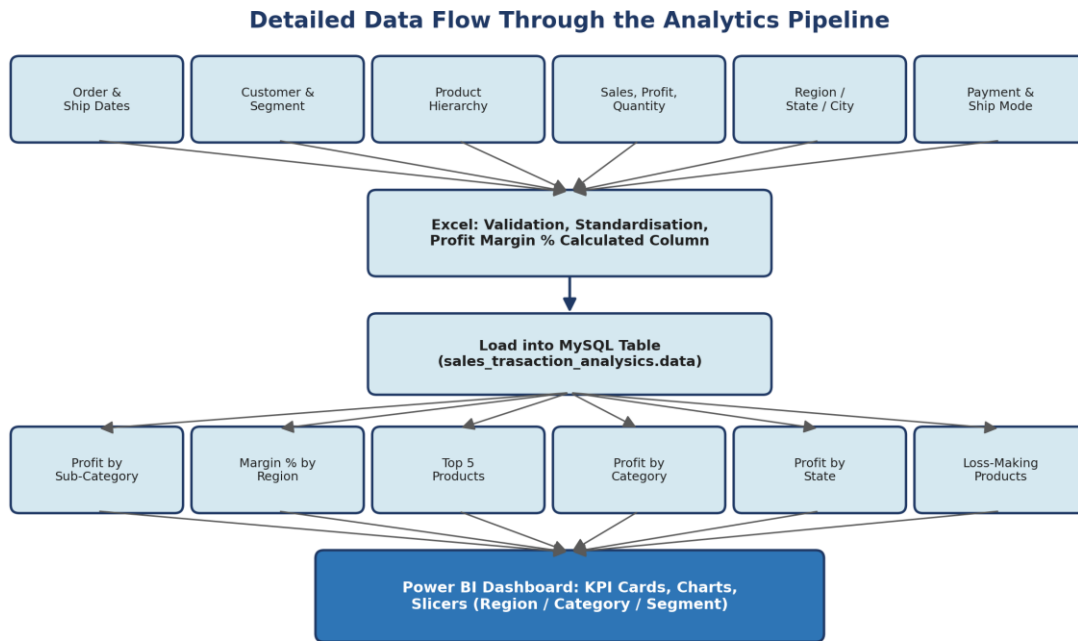


Figure 3.23: Detailed data flow Through the Analytics Pipeline

Figure 3.8 provides a more detailed representation of the data flow process. It illustrates how individual transaction-level attributes such as order information, customer details, product hierarchy, sales values, profit figures, geographical data, payment modes, and shipping details are integrated into a single cleaned dataset before being loaded into MySQL.

From the cleaned database table, the data is distributed into multiple SQL queries including Profit by Sub-Category, Profit Margin by Region, Top Products Analysis, Profit by Category, Profit by State, and Loss-Making Product Analysis. Each query addresses a specific business objective and generates meaningful analytical outputs.

The outputs generated by these SQL queries are then consolidated within the Power BI dashboard into KPI cards, charts, maps, and slicers. This fan-out and fan-in data flow architecture reflects a typical business intelligence workflow where raw data is transformed into structured information and ultimately converted into actionable business insights.

CHAPTER 4: RESULTS AND DISCUSSION

This chapter presents the key results obtained from the SQL queries described in Chapter 3, supported by the corresponding visualisations developed during the Power BI dashboard design. Each section presents a table or chart followed by a discussion of its business implications.

4.1 OVERALL BUSINESS PERFORMANCE

An overall business performance analysis was conducted to evaluate the company's sales, profitability, order volume, and operational efficiency during the period from 2019 to 2020. The purpose of this analysis was to obtain a high-level understanding of business performance and identify important trends that influence decision-making.

The following key performance indicators were analyzed:

- Total Sales generated across all transactions.
- Total Profit earned during the analysis period.
- Overall Profit Margin Percentage.
- Total number of orders processed.
- Total transaction line items recorded.
- Average shipping duration across orders.
- Number of returned items and return percentage.

The analysis included 5,901 transaction line items and 3,003 unique orders recorded between 2019 and 2020. The business generated total sales of approximately USD 1,565,804.32 and total profit of approximately USD 152,834.11, resulting in an overall profit margin of 9.76%. The average shipping duration across all orders was approximately 3.93 days.

Metric	Value
Total Sales	USD 1,565,804.32
Total Profit	USD 152,834.11
Overall Profit Margin	9.76%
Total Orders	3,003
Total Line Items	5,901
Average Shipping Duration	3.93 days
Returned Items	287 (4.86% of line items)

Table 4.1: Overall Business Performance Metrics

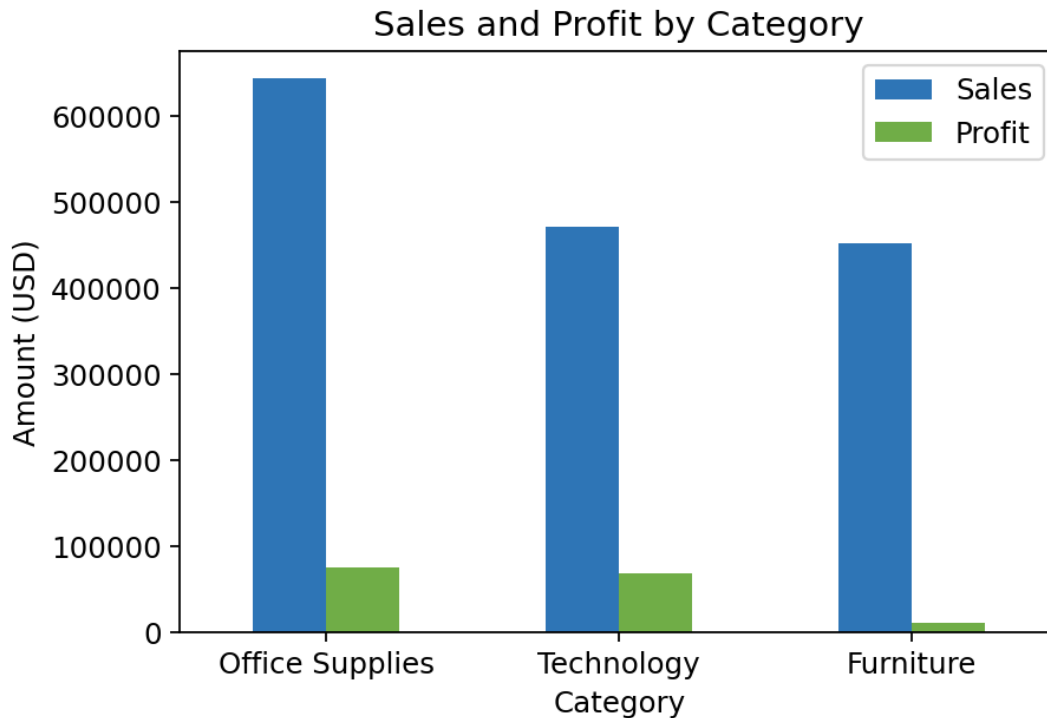


Figure 4.1: Sales and Profit by Product Category

The category-level analysis shows the contribution of each product category toward total sales and profit generation.

- Office Supplies generated the highest sales value of USD 643,707.69.
- Office Supplies also produced the highest profit of USD 74,797.25.
- Technology generated sales of USD 470,587.99 and profit of USD 68,030.25.
- Technology demonstrated a comparatively stronger profit contribution relative to sales volume.
- Furniture generated sales of USD 451,508.65.
- Furniture produced comparatively lower profit of USD 10,006.61 despite high sales volume.

The results indicate that Office Supplies and Technology performed strongly in terms of profitability, whereas Furniture displayed lower profit margins despite significant sales activity. This suggests that factors such as pricing strategies, discount policies, or operational costs may be affecting category-level profitability.

Overall, the business showed stable sales performance and positive profitability throughout the analysis period. The findings provide useful insights for improving category performance and supporting strategic business decisions.

4.2 PROFIT BY SUB-CATEGORY

SQL Query 1 (Section 3.3.1) was used to analyze profit at the sub-category level by aggregating the profit values of individual products. This analysis was performed to identify the most profitable and least profitable sub-categories and to understand their contribution to overall business performance.

The following observations were obtained from the sub-category analysis:

- Identification of highly profitable product sub-categories.
- Identification of loss-making product sub-categories.
- Comparison of profit performance across all sub-categories.
- Evaluation of category-level contribution toward overall profitability.
- Analysis of factors affecting product profit margins.
- Identification of products requiring improvement strategies.

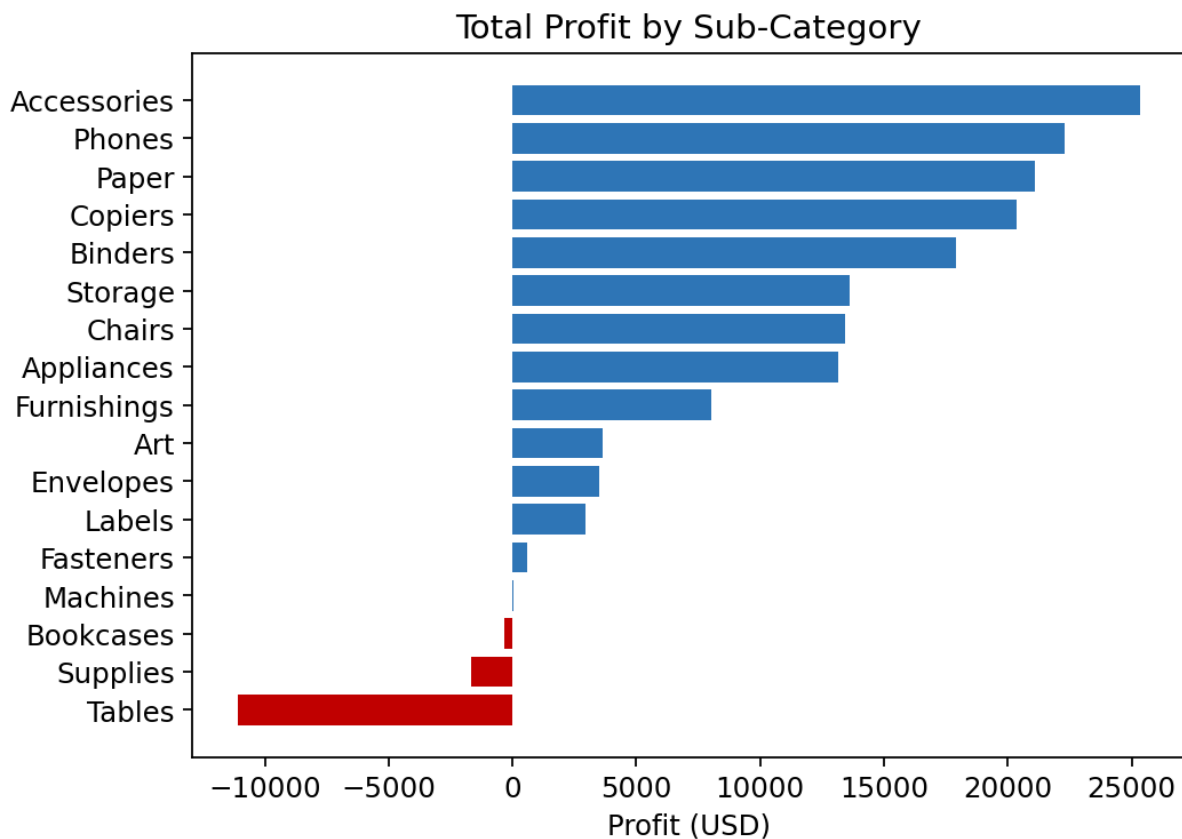


Figure 4.2: Total Profit by Sub-Category (SQL Query 1)

Result Grid	
Sub-Category	Total_Profit
Accessories	25336.645499999988
Phones	22250.492599999999
Copiers	20346.582800000004
Paper	20166.507999999999
Binders	17741.0164
Storage	13738.228900000002
Chairs	13406.703199999994
Appliances	13166.609799999997
Furnishings	7167.380599999998
Art	3635.9257000000002
Envelopes	3178.1756999999999
Labels	2937.2212000000002
Fasteners	596.08629999999997

Table 4.2: Profit by Sub-Category

The analysis indicates that Accessories, Phones, Paper, and Copiers were the highest profit-generating sub-categories. These product groups contributed significantly to total business profit and demonstrated strong performance within the overall product portfolio.

Among these, Accessories and Phones showed strong customer demand and positive profitability, while Paper and Copiers also generated substantial profit values. Their higher profitability may be influenced by factors such as strong sales performance, efficient pricing strategies, and favorable profit margins.

In contrast, Tables emerged as the largest loss-making sub-category, recording a total loss of USD 11,091.64. Supplies also showed a loss of USD 1,654.28, while Bookcases generated a loss of USD 342.89.

An important observation is that both Tables and Bookcases belong to the Furniture category. This finding explains the comparatively lower profitability of the Furniture category discussed in Section 4.1. The lower profit values may be associated with factors such as excessive discounting, higher transportation and handling costs, lower profit margins, and operational expenses related to bulky products.

The results suggest that loss-making sub-categories require further investigation and optimization strategies. Businesses can review pricing policies, discount structures, inventory management practices, and logistics costs to improve profitability and overall business performance.

4.3 PROFIT MARGIN % BY REGION

SQL Query 2 was used to calculate the Profit Margin Percentage for each sales region by dividing the total profit generated by total sales and converting the result into percentage values. This analysis was performed to evaluate regional performance and identify areas contributing positively or negatively to business profitability.

The following observations were obtained from the regional analysis:

- Comparison of sales performance across different regions.
- Calculation of profit margin percentages for each region.
- Identification of high-performing and low-performing regions.
- Analysis of the relationship between sales volume and profitability.
- Identification of regions requiring operational improvements.
- Evaluation of regional contribution toward overall business performance.

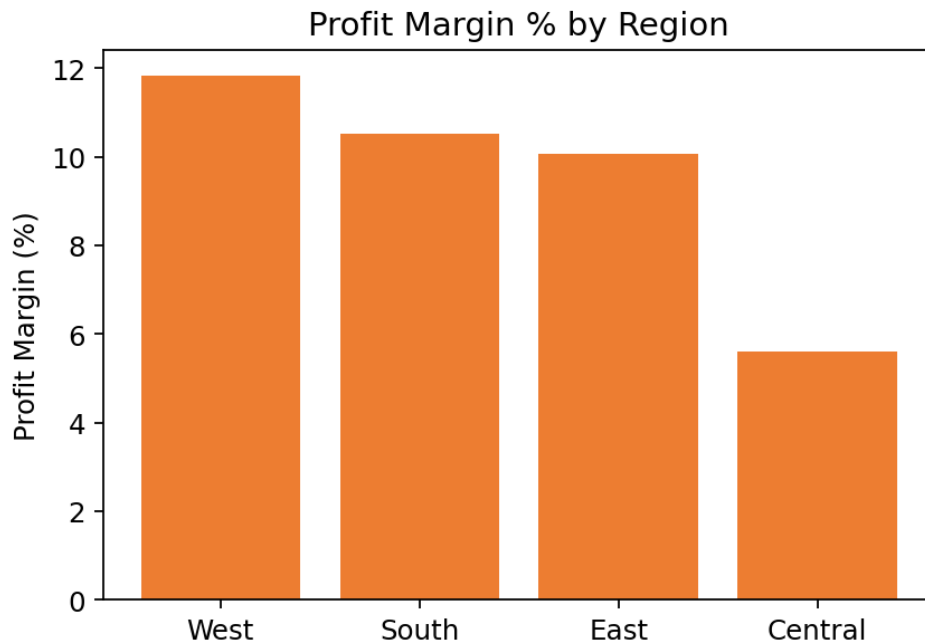


Figure 4.3: Profit Margin % by Region (SQL Query 2)

Result Grid			Filter Rows:
	Region	Profit_Margin_Percentage	
▶	West	11.77	
	South	10.56	
	East	10.11	
	Central	5.82	

Table 4.3: Profit Margin % by Region

The analysis shows that the West region recorded the highest sales value of USD 522,441.05 and also achieved the highest profit margin percentage of 11.77%. This indicates strong business performance and efficient profitability management within the region.

The West region emerged as the strongest-performing region due to its ability to generate high revenue while maintaining a comparatively higher profit margin. This suggests effective pricing strategies, customer demand, and better operational efficiency.

The Central region generated significant sales of approximately USD 341,007.52; however, its profit margin percentage was only 5.61%, which is considerably lower than that of the West region.

The lower profit margin despite substantial sales indicates that certain factors may be reducing profitability in the Central region. Possible reasons may include excessive discounting practices, higher transportation costs, increased operational expenses, or lower product profit margins.

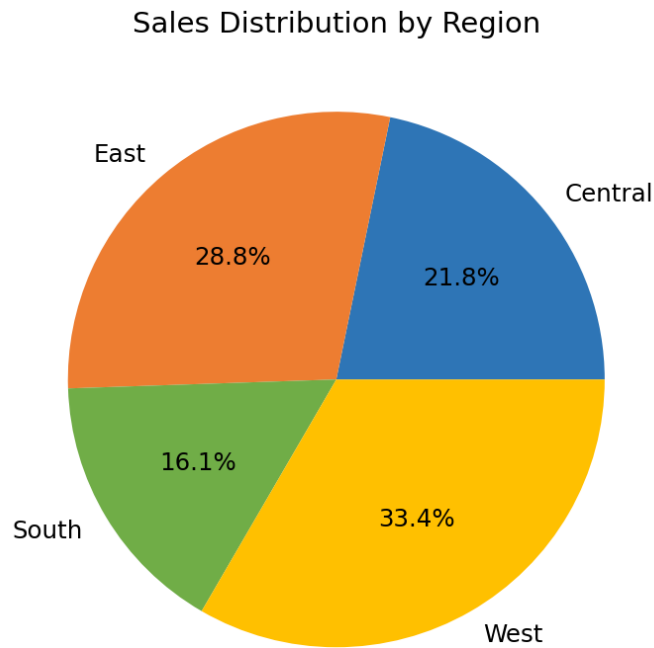


Figure 4.4: Share of Total Sales by Region

The regional sales distribution further illustrates the contribution of each region toward total business sales. The analysis indicates that some regions contribute higher sales volume while others contribute lower percentages of total revenue.

The findings suggest that regions with lower profit margins require further investigation and improvement strategies. Organizations may review pricing policies, cost structures, promotional activities, and regional business operations to improve profitability and enhance overall performance.

Overall, regional analysis provides valuable insights for understanding geographical business performance and supports effective decision-making for future growth and resource allocation.

4.4 TOP 5 PRODUCTS BY PROFIT

SQL Query 3 was used to rank individual products according to their total profit contribution. This analysis was performed to identify the products generating the highest profit and to understand their impact on overall business performance.

The following observations were obtained from the product-level analysis:

- Identification of the highest profit-generating products.
- Comparison of product performance based on profit values.
- Analysis of product contribution toward total business profit.
- Evaluation of relationships between sales and profitability.
- Identification of premium products with stronger profit margins.
- Understanding customer demand and product performance trends.

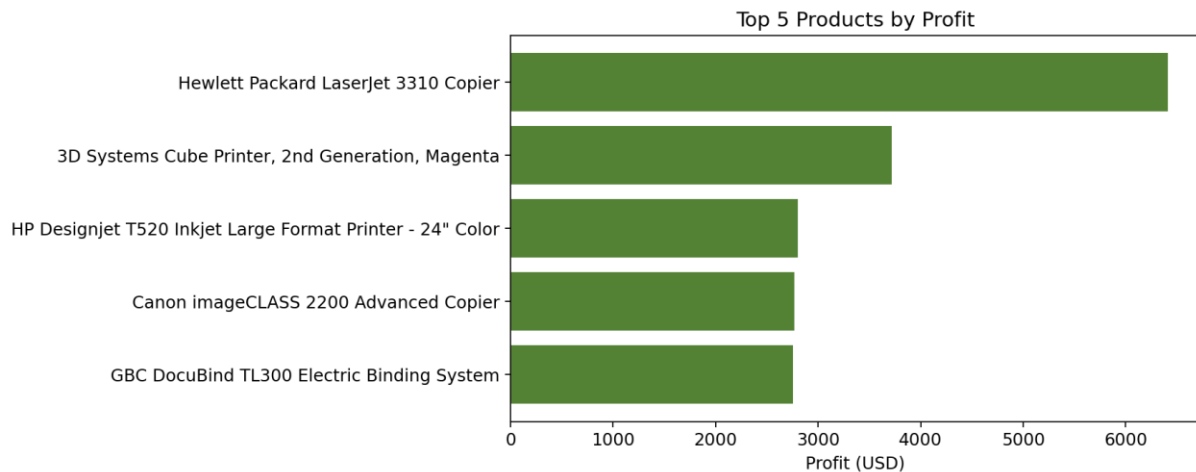


Figure 4.5: Top 5 Products by Profit (SQL Query 3)

Product Name	Total_Sales	Total_Profit	Profit_Margin_Percentage
Hewlett Packard LaserJet 3310 Copier	13837.732	6407.8932	46.31
3D Systems Cube Printer, 2nd Generation, Mag...	14334.89	3717.9714000000004	25.94
HP Designjet T520 Inkjet Large Format Printer ~...	4749.95	2799.984	58.95
Canon imageCLASS 2200 Advanced Copier	14076.824	2771.928	19.69
GBC DocuBind TL300 Electric Binding System	12890.258	2753.7592999999997	21.36

Table 4.4: Top 5 Products by Profit

The analysis revealed that the five most profitable products in the dataset belonged to the Technology category. These products contributed significantly toward the overall profit generated by the business and demonstrated strong market performance.

Among the identified products, the HP Design jet T520 Inkjet Large Format Printer achieved the highest profit margin of approximately 58.95%, despite having comparatively lower sales values than some other top-performing products.

This observation indicates that high sales volume does not always result in maximum profitability. Certain specialized and premium products can generate higher profit margins even with fewer sales transactions.

The analysis also suggests that products related to printing and copying equipment provide stronger profitability compared to regular high-volume products. Their higher margins may be associated with specialized features, premium pricing strategies, and lower competition.

The findings provide valuable business insights that can support product planning, inventory management, and marketing strategies. Organizations can focus on promoting high-margin products while maintaining balanced sales performance across other product categories.

Overall, product-level profit analysis helps identify the products that contribute most significantly to business profitability and supports effective decision-making for future business growth.

4.5 MOST LOSS-MAKING PRODUCTS

SQL Query 4 was used to identify products whose cumulative profit across all transactions was negative. This analysis was performed to determine products that negatively impacted overall business profitability and to identify areas requiring improvement.

The following observations were obtained from the loss-making product analysis:

- Identification of products with negative cumulative profit values.
- Analysis of product-level profitability performance.
- Detection of products contributing to business losses.
- Comparison of loss values among different products.
- Identification of high-cost products with poor profit performance.
- Evaluation of the impact of pricing and discount strategies on profitability.
- Understanding the relationship between sales volume and product profit.

Product Name	Total_Profit
Cubify CubeX 3D Printer Double Head Print	-6239.9792
Cubify CubeX 3D Printer Triple Head Print	-3839.9904
Lexmark MX611dhe Monochrome Laser Printer	-2719.984
Bush Advantage Collection Racetrack Conferen...	-2019.2395999999999
Ibico EPK-21 Electric Binding System	-1285.1932
Martin Yale Chadless Opener Electric Letter Ope...	-1232.5588
BoxOffice By Design Rectangular and Half-Moo...	-986.5625
Bretford "Just In Time" Height-Adjustable Multi-...	-964.194
Zebra GK420t Direct Thermal/Thermal Transfer ...	-938.28
Epson TM-T88V Direct Thermal Printer - Monoch...	-935.9595
Chromcraft 48" x 96" Racetrack Double Pedest...	-897.7919999999999
Chromcraft Bull-Nose Wood Oval Conference T...	-870.5484
Bevis Oval Conference Table, Walnut	-751.6224

Table 4.5: Most Loss-Making Products

The analysis showed that a total of 310 distinct products recorded a net loss during the observed period. Among the identified products, the Cubify CubeX 3D Printer in both double-head and triple-head variants represented the largest loss-making products in the dataset, together contributing losses exceeding USD 10,000.

Additionally, furniture products such as conference and meeting tables also appeared among the highest loss-generating products. This indicates that bulky and high-cost products contributed significantly to overall losses.

Several factors may be responsible for these negative profit values, including excessive discounts, high transportation and handling costs, lower customer demand, and reduced profit margins associated with expensive products.

The findings suggest that products generating continuous losses require detailed business evaluation. Organizations may consider revising pricing policies, controlling discount rates, optimizing logistics costs, and reviewing product strategies to improve profitability.

Overall, this analysis provides valuable insights into products that negatively affect business performance and helps support data-driven decision-making for improving operational efficiency and financial outcomes.

4.6 PROFIT BY STATE

SQL Query 5 was used to analyze profit at the state level by aggregating the profit values for each state. The resulting visualization displays the top five profit-generating states and the bottom five loss-making states.

The purpose of this analysis was to understand the geographical distribution of profit and identify states that contribute positively or negatively to the overall business performance. State-wise profit analysis helps organizations evaluate market performance, identify profitable regions, and detect areas that may require improvement strategies.

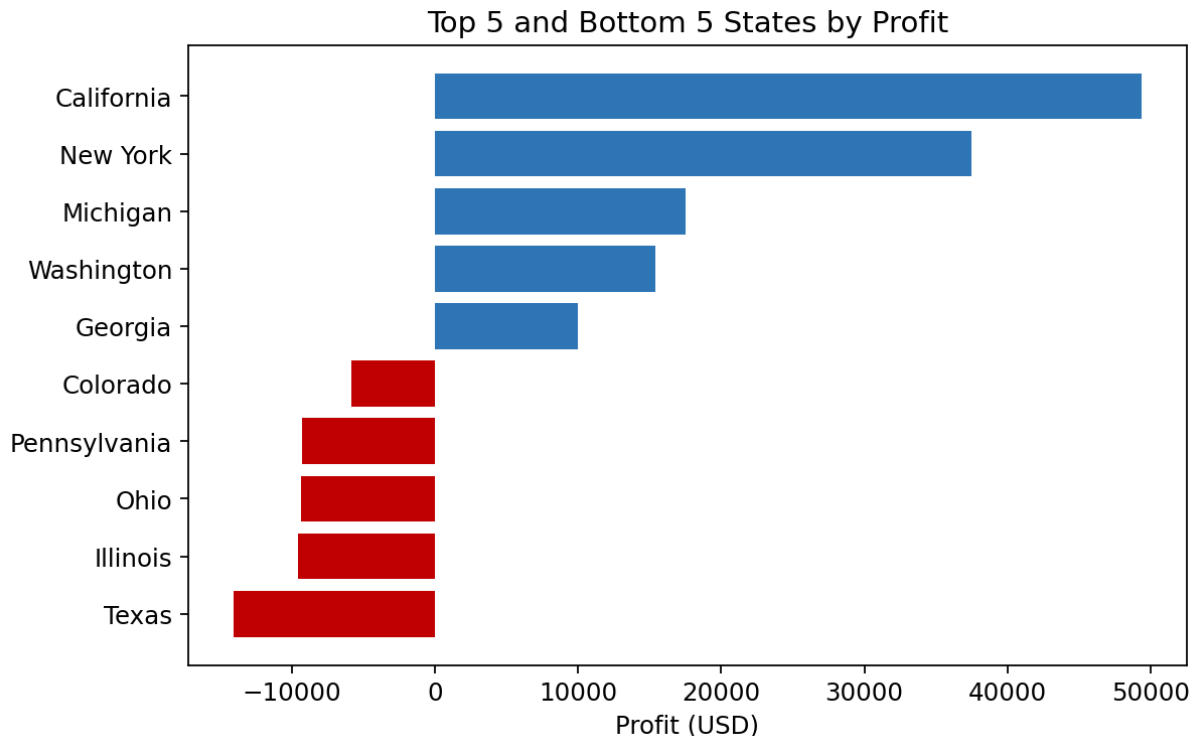
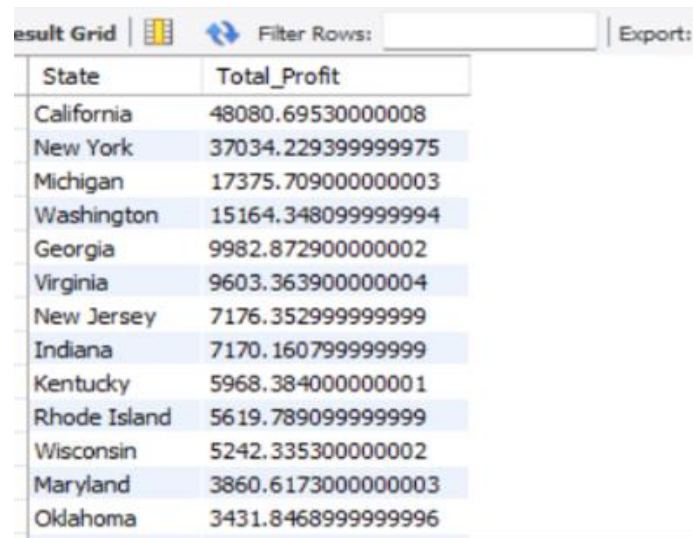


Figure 4.6: Top 5 and Bottom 5 States by Profit (SQL Query 5)



State	Total_Profit
California	48080.695300000008
New York	37034.229399999975
Michigan	17375.709000000003
Washington	15164.348099999994
Georgia	9982.872900000002
Virginia	9603.363900000004
New Jersey	7176.352999999999
Indiana	7170.160799999999
Kentucky	5968.384000000001
Rhode Island	5619.789099999999
Wisconsin	5242.335300000002
Maryland	3860.6173000000003
Oklahoma	3431.8468999999996

Table 4.6: Profit by State (Top 5 / Bottom 5)

The analysis indicates that California and New York emerged as the highest profit-generating states, contributing a significant share of the overall profit among the top-performing states. Their strong performance reflects high sales volume, customer demand, and effective business operations within these regions.

The remaining top-performing states also demonstrated positive contributions toward total profit, indicating stable business performance and efficient sales management. Strong profitability in these states may be associated with higher customer purchasing activity and better product performance.

In contrast, Texas recorded the highest cumulative loss with a total loss of USD 14,078.16, followed by Illinois, Ohio, and Pennsylvania. These states showed negative profit values, indicating lower profitability despite maintaining sales activity.

Several factors may contribute to losses in these regions, including excessive discounting, higher shipping costs, lower profit margins, or ineffective pricing strategies. A detailed investigation into operational and marketing practices in these states may help identify the root causes of poor performance.

The results suggest that the loss-making states require focused managerial attention and corrective actions related to pricing policies, discount strategies, transportation costs, and regional business planning. Addressing these issues may improve profitability and overall business performance.

4.7 SALES AND PROFIT BY CUSTOMER SEGMENT

Segment	Total Sales (USD)	Total Profit (USD)
Consumer	753,002.13	70,754.59
Corporate	509,743.13	49,489.80
Home Office	303,059.07	32,589.72

Table 4.7: Sales and Profit by Customer Segment

SQL analysis was performed to examine the contribution of different customer segments toward overall sales and profit. The dataset was grouped into three customer segments: Consumer, Corporate, and Home Office. This analysis helps in understanding customer purchasing behavior and identifying the segments that contribute the most to business growth and profitability.

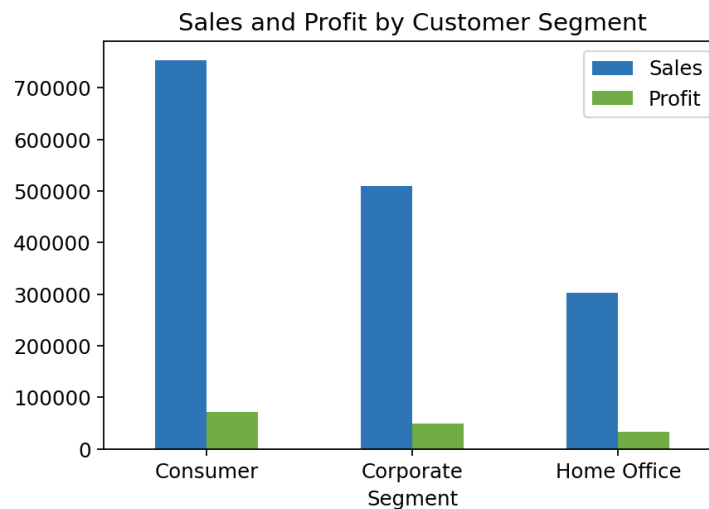


Figure 4.7: Sales and Profit by Customer Segment

The analysis shows that the Consumer segment is the largest contributor to both sales and profit, generating the highest revenue among all customer groups. This indicates that individual consumers represent the major customer base of the organization and play a significant role in overall business performance.

The Corporate segment ranks second in both sales and profit generation. This suggests that business organizations and corporate customers also contribute substantially to revenue, although their overall contribution remains lower than that of the Consumer segment.

The Home Office segment generated the lowest sales and profit values among the three customer groups. Although this segment contributes less compared to Consumer and Corporate customers, it still provides a positive contribution toward overall company performance.

The ranking pattern of the customer segments remains consistent for both sales and profit, with Consumer leading, followed by Corporate and Home Office. This indicates that profit generation across customer segments is closely associated with sales volume and customer purchasing activity rather than significant differences in profit margins.

The findings suggest that the organization should continue focusing on the Consumer segment while also exploring opportunities to strengthen engagement and sales strategies for Corporate and Home Office customers to increase overall business growth.

4.8 OPERATIONAL INSIGHTS: Payment Mode and Shipping

Operational analysis was performed to understand customer payment preferences and shipping behavior. The purpose of this analysis was to identify commonly used payment

methods and delivery patterns that may influence customer satisfaction and operational efficiency.

The following operational insights were obtained from the analysis:

- Analysis of customer payment preferences across different payment modes.
- Identification of the most frequently used payment method based on total sales value.
- Comparison of Cash on Delivery, Online Payment, and Card Payment usage.
- Evaluation of customer dependence on digital and traditional payment methods.
- Analysis of shipping methods used for product delivery.
- Identification of the most commonly selected shipping mode by customers.
- Examination of average delivery duration across transactions.
- Comparison between standard and expedited shipping services.
- Understanding the impact of shipping preferences on customer behavior.
- Identification of opportunities to improve operational efficiency and customer experience.

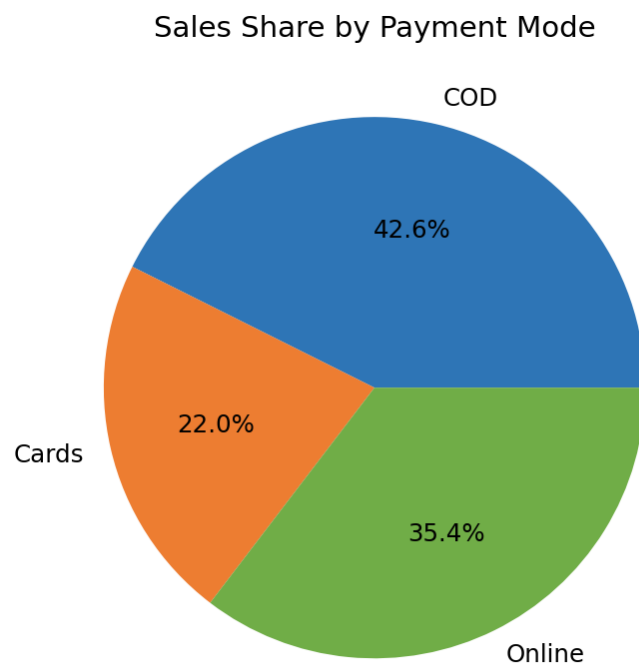


Figure 4.8: Sales Share by Payment Mode

The analysis of payment modes revealed that Cash on Delivery (COD) was the most frequently used payment method based on total sales value, generating USD 667,417.75. Online payments ranked second with total sales of USD 553,993.46, while Card payments accounted for USD 344,393.11.

The higher usage of Cash on Delivery indicates that a large proportion of customers prefer paying at the time of product delivery rather than making payments in advance. Customer

preference for COD may be influenced by factors such as trust, convenience, and payment flexibility. However, heavy dependence on COD can lead to additional operational challenges including delayed payment confirmation, increased return rates, and higher cash-handling risks.

The findings suggest that businesses can encourage customers to adopt digital payment methods by offering discounts, cashback rewards, loyalty points, or promotional benefits. Increasing the adoption of online payment methods may improve transaction efficiency and reduce operational risks.

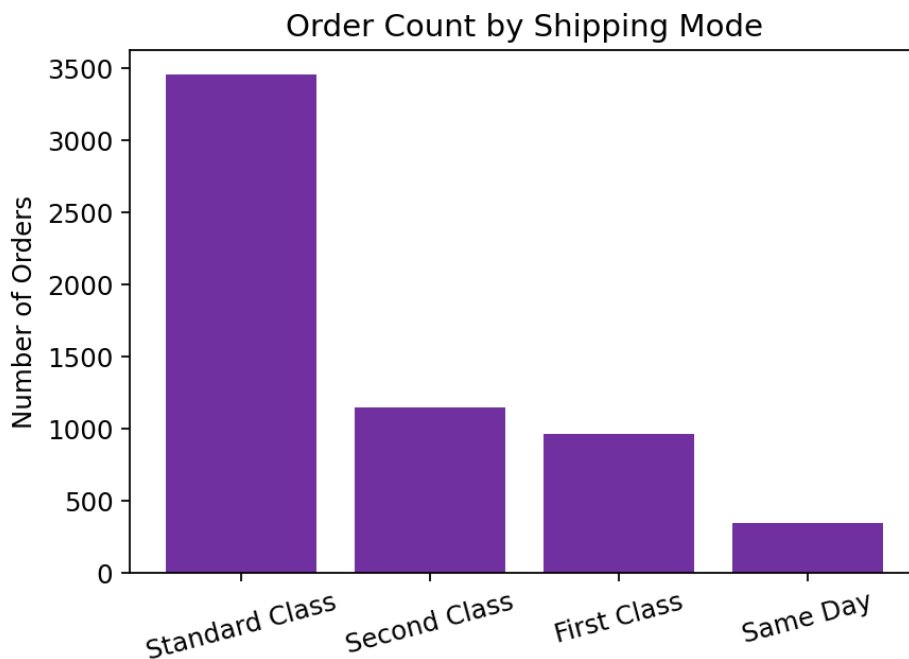


Figure 4.9: Order Count by Shipping Mode

Standard Class was the dominant shipping mode by order count, consistent with an average shipping duration of 3.93 days observed across the dataset. The relatively low usage of Same Day and First Class shipping suggests limited current demand for expedited delivery options, though this could also reflect pricing or availability constraints that may merit further investigation.

The shipping analysis indicated that Standard Class was the most commonly used shipping method based on order count. This suggests that customers generally prefer regular delivery services due to lower shipping costs and acceptable delivery times.

The dataset also showed an average shipping duration of approximately 3.93 days, indicating relatively efficient delivery operations. Shipping modes such as First Class and Same Day delivery recorded lower usage compared to Standard Class.

The limited adoption of faster shipping methods may be influenced by factors such as additional shipping charges, customer preferences, or service availability constraints. Further analysis can be conducted to determine whether promotional offers or optimized logistics strategies could increase customer interest in expedited delivery options.

Overall, the operational insights obtained from payment and shipping analysis provide valuable information for improving customer experience, optimizing business operations, and supporting strategic decision-making.

4.9 MONTHLY SALES AND PROFIT TREND

Monthly trend analysis was performed to understand changes in sales and profit over time and to identify recurring business patterns across different months. This analysis helps in evaluating business growth, detecting seasonal trends, and understanding variations in customer purchasing behavior.

The following observations were derived from the monthly trend analysis:

- Analysis of monthly sales performance across the observed period.
- Identification of profit trends over different months.
- Comparison between sales growth and profit growth patterns.
- Detection of seasonal fluctuations in customer purchasing behavior.
- Identification of peak sales periods during specific months.
- Analysis of low-performing periods with reduced sales activity.
- Evaluation of profit changes corresponding to sales variations.
- Understanding the relationship between sales volume and profitability.

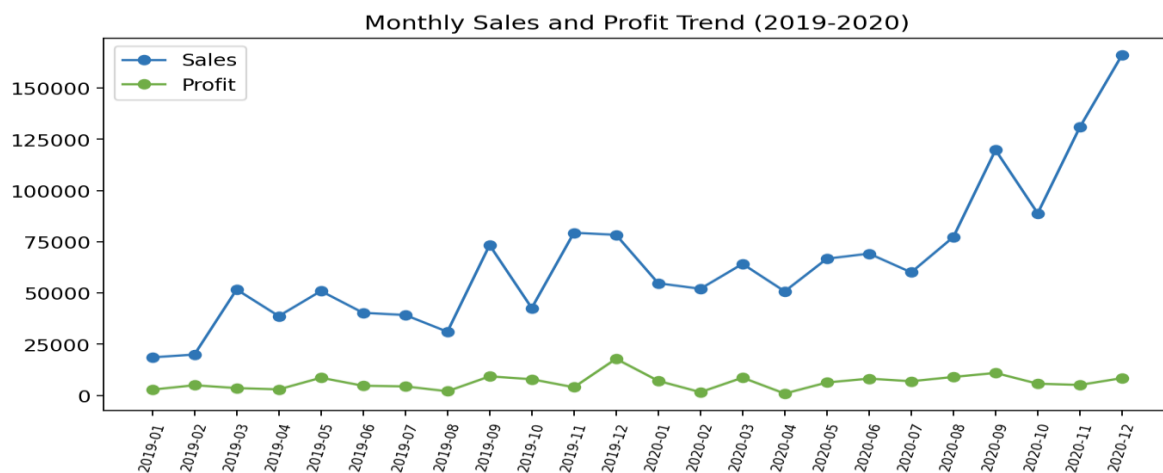


Figure 4.10: Monthly Sales and Profit Trend, 2019–2020

The monthly trend chart, which formed the centrepiece of the Power BI dashboard’s overview page, shows clear seasonal variation in sales across the two-year period, with recurring peaks toward the final quarter of each calendar year — a pattern consistent with year-end and holiday-season purchasing behaviour commonly observed in retail and office-supply businesses. Profit follows the same broad seasonal pattern as sales but at a visibly compressed scale, reflecting the modest overall profit margin of 9.76% computed in Section 4.1.

The monthly trend chart, which formed a major component of the Power BI dashboard overview page, shows noticeable seasonal variations in sales and profit throughout the two-year period. The analysis indicates recurring increases in sales during the final quarter of each calendar year, reflecting higher customer purchasing activity commonly associated with year-end periods and seasonal demand.

Profit follows a similar trend to sales but at a comparatively lower scale, indicating that although revenue increases during high-demand periods, profit growth remains moderate due to factors such as discounts, operational costs, and overall profit margins.

The observed trend provides useful business insights for planning inventory management, promotional activities, and resource allocation during high-demand and low-demand periods. Understanding these patterns supports better forecasting and improved decision-making for future business operations.

4.10 POWER BI DASHBOARD INTERACTIVE FEATURES

The results obtained from SQL analysis and business insights were integrated into an interactive Power BI dashboard. The dashboard was designed to provide users with a visual and user-friendly interface for exploring sales and profitability data. Two dashboard pages were developed: the Sales Transaction Dashboard and the Profit Transaction Dashboard.

The Sales Transaction Dashboard focuses on overall business performance and includes important metrics and visual representations related to sales activity. The dashboard contains the following features:

- KPI cards displaying Total Sales, Orders, Profit, Profit Margin Percentage, and Quantity Sold.
- Sales by Sub-Category bar chart showing product category performance.
- Monthly Sales Trend chart for analyzing sales fluctuations over time.
- Sales by Segment donut chart for customer segment analysis.
- Sales by State map visualization showing geographical sales distribution.
- Interactive filters including Order Date, Segment, Category, and Region slicers.

The Profit Transaction Dashboard focuses on profitability analysis and business performance evaluation. The dashboard includes:

- KPI cards displaying Profit, Profit Margin Percentage, Average Profit per Order, and Loss-Making Orders.
- Profit by Sub-Category chart identifying profitable and loss-making products.
- Monthly Profit Trend chart showing profit fluctuations.
- Profit Margin Percentage by Region chart for regional comparison.
- Top 5 Products by Profit table displaying high-performing products.
- Interactive slicers for filtering dashboard results dynamically.

The dashboards provide interactive functionality where selecting filters such as Region, Category, Segment, or Date automatically updates all charts and KPI values on the page. This interactivity allows users to explore business performance dynamically without writing SQL queries.

Overall, the Power BI dashboard transformed analytical outputs into visually understandable insights and supported faster and more effective business decision-making.

1. KPI Cards

- Display important business metrics.

- Show Total Sales, Orders, Profit, Profit Margin %, and Quantity Sold.
 - Provide a quick overview of business performance.
2. **Sales by Sub-Category Chart**
 - Displays sales values for different product sub-categories.
 - Helps identify top-selling and low-selling products.
 - Supports product performance analysis.
 3. **Sales Trend Chart**
 - Shows monthly sales performance over time.
 - Helps identify growth patterns and seasonal trends.
 - Supports future sales forecasting.
 4. **Sales by Segment Chart**
 - Represents sales contribution by customer segments.
 - Includes Consumer, Corporate, and Home Office categories.
 - Helps understand customer purchasing behavior.
 5. **Sales by State Map**
 - Displays geographical sales distribution across states.
 - Helps identify high-performing and low-performing regions.
 - Supports regional business analysis.
 6. **Profit by Sub-Category Chart**
 - Shows profit generated by each product sub-category.
 - Identifies profitable and loss-making products.
 - Helps improve product strategies.
 7. **Profit Trend Chart**
 - Displays monthly profit changes over time.
 - Helps monitor profit growth patterns.
 - Supports business performance evaluation.
 8. **Profit Margin by Region Chart**
 - Compares profit margins across different regions.
 - Identifies regions with better profitability.
 - Supports regional decision-making.
 9. **Top 5 Products by Profit Table**
 - Displays products generating maximum profit.
 - Includes sales and profit information.
 - Helps identify best-performing products.

10. Slicers and Filters

- Used for dynamic dashboard interaction.
- Includes Order Date, Category, Segment, and Region filters.
- Automatically updates all charts and KPI values.

CHAPTER 5: CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

The semester industrial training undertaken on the Sales Transaction Analysis project provided comprehensive, hands-on exposure to the complete data analytics workflow used in industry — from raw data cleaning to interactive dashboard delivery. Working with a dataset of 5,901 transaction records and 3,003 orders spanning 2019–2020, the training combined Microsoft Excel for data cleaning and exploratory analysis, SQL for precise aggregation and filtering, and Microsoft Power BI for interactive visualisation.

The analysis established that the business achieved total sales of approximately USD 1.57 million and total profit of approximately USD 152,834, an overall profit margin of 9.76%. Office Supplies was identified as the most profitable category, while Furniture — driven specifically by the Tables sub-category — was found to be a net drag on profitability. The West region delivered both the highest sales and the highest profit margin (11.83%), while the Central region lagged considerably at 5.61%. At the product level, premium printing and copying equipment generated the highest profit margins, while large 3D printers and bulky conference tables were identified among the most significant loss-making products. At the state level, Texas, Illinois, Ohio and Pennsylvania were identified as the principal loss-making states requiring managerial attention.

Beyond the specific business findings, the training strengthened the trainee's practical skills in writing GROUP BY, HAVING and aggregate SQL queries, in building PivotTable-based exploratory summaries in Excel, and in designing a multi-page, slicer-driven Power BI dashboard — skills that are directly transferable to the role of a Data Analyst or Business Intelligence developer in industry.

5.2 BUSINESS RECOMMENDATIONS

- Review the pricing and discounting strategy for the Tables sub-category and for large-format equipment such as 3D printers, since these consistently appear among the most significant sources of loss.
- Investigate the cost structure and discounting practices in the Central region, given its profit margin is roughly half that of the best-performing West region despite comparable sales volume.
- Conduct a focused review of sales operations in Texas, Illinois, Ohio and Pennsylvania, the four states with the largest net losses.
- Consider promotional incentives to shift customers from Cash on Delivery toward Card or Online payment modes, to reduce payment-processing risk and improve cash-flow predictability.
- Leverage the strong, consistent margins of the Accessories, Phones, Paper and Copiers sub-categories by prioritising inventory and marketing investment toward these segments.

5.3 FUTURE SCOPE

Although the current project successfully achieved the objectives of analyzing historical sales transaction data and developing an interactive business intelligence dashboard, several enhancements and advanced analytical techniques can be incorporated in future developments to increase the effectiveness and practical value of the system.

The possible future improvements are as follows:

1. Predictive Analytics

- Apply machine learning and time-series forecasting models such as ARIMA and Exponential Smoothing.
- Predict future sales and profit trends using historical transaction data.
- Support business planning and demand forecasting.
- Improve inventory and resource management decisions.

2. Customer Segmentation

- Implement clustering techniques such as K-Means clustering for customer grouping.
- Segment customers according to purchasing behavior and spending patterns.
- Identify high-value and low-value customers.
- Support personalized marketing and customer retention strategies.

3. Discount and Pricing Optimization

- Analyze the relationship between discount percentages and profitability.
- Identify the impact of discount policies on sales performance.
- Optimize pricing strategies for products with lower profit margins.
- Reduce losses from excessive discounting practices.

4. Automated Data Refresh and Real-Time Integration

- Connect Power BI directly with a live transactional database.
- Enable automatic dashboard updates when new records are added.
- Reduce manual data import and processing activities.
- Support real-time business monitoring and reporting.

5. Return and Loss Analysis

- Conduct detailed analysis of returned products and customer returns.
- Identify products with higher return rates.
- Analyze regional and shipping-related factors affecting returns.
- Improve product quality and customer satisfaction.

6. Advanced Dashboard Features

- Add drill-through reports and advanced filtering options.
- Include AI-based visual insights and anomaly detection features.

- Enhance dashboard usability and interactive capabilities.
- Provide more detailed analytical reports for stakeholders.

7. Integration of Machine Learning Models

- Integrate predictive machine learning algorithms within the analytics pipeline.
- Generate recommendations for products and customer preferences.
- Support intelligent decision-making processes.
- Improve business forecasting accuracy.

These future enhancements would extend the current descriptive and diagnostic analytics system toward predictive and prescriptive analytics. Such improvements can help organizations make more intelligent, data-driven, and proactive business decisions.

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Microsoft Learn Platform<https://learn.microsoft.com/>

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- **Data Analytics Learning Resources**
Analytics Vidhya<https://www.analyticsvidhya.com/>

APPENDIX

APPENDIX A: COMPLETE SQL QUERY CODE

The complete set of SQL queries written and executed during the training is reproduced below for reference.

A.1 Base Table Query

```
SELECT * FROM connect.sales_trasaction_analysis;
```

A.2 Profit by Sub-Category

```
SELECT  
  
    `Sub-Category`, SUM(Profit) AS Total_Profit  
  
FROM  
  
    connect.sales_trasaction_analysis  
  
GROUP BY `Sub-Category`  
  
ORDER BY Total_Profit DESC;
```

A.3 Profit Margin % by Region

```
SELECT  
  
    Region,  
  
    ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Profit_Margin_Percentage  
  
FROM  
  
    connect.sales_trasaction_analysis  
  
GROUP BY Region  
  
ORDER BY Profit_Margin_Percentage DESC;
```

A.4 Top 5 Products by Profit

```
SELECT  
  
    `Product Name`,  
  
    SUM(Sales) AS Total_Sales,  
  
    SUM(Profit) AS Total_Profit,  
  
    ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Profit_Margin_Percentage  
  
FROM  
  
    connect.sales_trasaction_analysis
```

GROUP BY `Product Name`

ORDER BY Total_Profit DESC

LIMIT 5;

A.5 Region Filter (Distinct Regions)

SELECT DISTINCT

Region

FROM

connect.sales_trasaction_analysis

ORDER BY Region;

A.6 Profit by Category

SELECT

Category, SUM(Profit) AS Total_Profit

FROM

connect.sales_trasaction_analysis

GROUP BY Category

ORDER BY Total_Profit DESC;

A.7 Profit by State

SELECT

State, SUM(Profit) AS Total_Profit

FROM

connect.sales_trasaction_analysis

GROUP BY State

ORDER BY Total_Profit DESC;

A.8 Most Loss-Making Products

SELECT

`Product Name`, SUM(Profit) AS Total_Profit

FROM

connect.sales_trasaction_analysis

GROUP BY `Product Name`

HAVING SUM(Profit) < 0

ORDER BY Total_Profit;

APPENDIX B: DATASET STRUCTURE

Field Name	Description
Row ID	Unique identifier for each transaction record
Order ID	Unique order identification number
Order Date	Date on which the order was placed
Ship Date	Date on which the order was shipped
Ship Mode	Shipping method used
Customer ID	Unique customer identification
Customer Name	Name of the customer
Segment	Customer category (Consumer, Corporate, Home Office)
Country	Customer country
City	Customer city
State	Customer state
Region	Business region
Product ID	Unique product identification
Category	Product category
Sub-Category	Product sub-category
Product Name	Name of product
Sales	Total sales amount
Quantity	Quantity of items sold
Profit	Profit earned
Profit Margin %	Profit percentage relative to sales
Returns	Indicates returned products
Payment Mode	Payment method used
Shipping Duration	Number of days between order and delivery
Cost of Goods	Product cost value
Revenue per Unit	Revenue generated per unit